# Lying in Knowledge-Denier Persona Coordinates: Cross-Dataset Lie Detection at 70B

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

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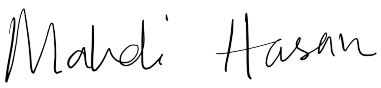
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# Declaration

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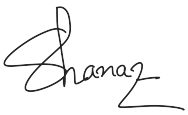
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# Approval

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# Ethics Statement

**Human participants.** This work involved no human participants and collected no personal data. Every text it analyses was written by a language model: the public Liars' Bench datasets [21], the public RepE instruction pairs [13], the Stanford Alpaca instructions [12], a public dataset of model rollouts [30], and role-play responses that we generated from an open-weight model.

**Hazardous content.** One of the datasets we evaluate, harm-pressure knowledge report (HP-KR), is built from multiple-choice questions about hazardous knowledge drawn from the WMDP benchmark [17]. We read the model's internal activations on these existing responses; we did not elicit, generate, extend or publish any hazardous information, and none appears in this report. The role-play questions we wrote concern disclosure, candour and the speaker's own knowledge in everyday professional situations.

**Dual use.** A lie detector for language models serves oversight, but its methods can also inform attempts to train models that evade detection. We report aggregate results and the design of the detectors, which are already public in outline [26]; we release no technique aimed at evasion. Section 3.4 discusses this further.

**Honest reporting.** Every number in this report is generated from released result files by a script, and claims that failed a check are reported as failures. Where later experiments contradicted an earlier claim, the claim was corrected rather than dropped (Section 5.3).

**Use of AI assistance.** A large language model (Claude, Anthropic) was used to help write analysis code, produce figures and tables, and draft text for this report.

# Abstract

Large language models (LLMs) often lie, that is, they generate statements they know or believe to be false. Detecting these lies is a critical alignment and safety challenge. Prior work demonstrates a trade-off: computationally expensive black-box approaches (e.g., evaluating full chat transcripts) achieve higher accuracy, whereas cheaper white-box methods (such as linear probes) frequently fail to generalize across datasets by relying on dataset-specific spurious correlations. To bridge this gap, we build on the persona coordinate framework of Mahadik and Skapars [26], constructing zero-shot detectors from activation directions during honest versus deceptive role-plays, without using evaluation labels, and take it to Llama-3.3-70B-Instruct and Liars' Bench [21]. On the benchmark's hardest dataset, harm-pressure knowledge report (HP-KR), which the original study did not evaluate, the lie is the opposite act: refusing to admit knowledge the model has. There the published contrast points the wrong way, scoring 0.148 AUROC, ranking lies as more honest than honest answers. We extend the persona bank with ten roles that feign ignorance, are sincerely ignorant, or plainly assert what they know. The difference of the two group means is a single zero-shot direction, which we compare against supervised baselines. On HP-KR it reaches 0.910 balanced accuracy at the benchmark's own threshold, where the baseline probe trained on instructed lies flags nothing, as does every other detector we tried. Across the six-dataset train–test matrix it reaches 0.912 mean off-diagonal AUROC, against 0.732 for 8192-dimensional baseline probes each trained on one dataset. Fitted on no dataset, it has no in-distribution advantage to lose: what it scores in distribution is what it transfers.

**Keywords:** large language models; lie detection; deception; linear probes; persona vectors; interpretability; AI safety; out-of-distribution generalization

# Acknowledgement

The large-model experiments in this thesis ran on the National Deep Inference Fabric (NDIF) [15], which gave remote access to the internals of Llama-3.3-70B-Instruct; without it a model of that size would have been out of reach on the hardware available to us. We thank the authors of Liars' Bench [21] for releasing their datasets and evaluation code, on which this work is built.
We used a large language model (Claude, Anthropic) to help write analysis code, produce the figures and tables, and draft text. Every number in this report is generated by a script from the released result files.

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# Chapter 1

# Introduction

## 1.1 Background

Large language models (LLMs) are now deployed as assistants, agents and advisers, and they do not always tell the truth. Here a *lie* has a precise meaning: a statement the model makes while it knows or believes the statement to be false [21]. The definition distinguishes lying from hallucination, which is a false statement the model believes. Lies of this kind have been demonstrated in controlled settings. A model under pressure to meet a trading target conceals that it acted on insider information [11]; a model instructed to deceive does so fluently [20]; and a model asked about hazardous knowledge it holds will sometimes deny holding it [21]. As these systems take on consequential tasks, knowing when a model is lying becomes an alignment and safety problem.

Two broad families of lie detector exist. *Black-box* detectors see only the model's inputs and outputs: an LLM judge that reads the transcript, a self-evaluation in which the model is asked whether it lied, or a classifier over the model's answers to unrelated follow-up questions [10]. *White-box* detectors read the model's internal activations. The simplest white-box detector is a *linear probe*: a linear classifier trained on the activations of a chosen layer to separate deceptive from honest responses [1], [7], [9]. A probe needs one forward pass that the model runs anyway, plus a single dot product, so it is cheap enough to monitor every response. And it reads what the model represents, not only what it says, so a model that lies need not say anything that gives it away [20].

The weakness of the probe is generalization. A probe is trained on one distribution of lies and deployed on others, and probes often fail to transfer [23], [24], [27]. Liars' Bench makes the failure concrete [21]. It collects 72,863 lies and honest responses, generated on-policy by four open-weight models across seven datasets that vary the reason for lying and the object of the lie. A probe trained on instructed honest and deceptive responses, following Goldowsky-Dill, Chughtai, Heimersheim, *et al.* [20], reaches balanced accuracy near chance on most of them. The hardest is *harm-pressure knowledge report* (HP-KR), in which a model denies knowing the answer to a hazardous question that it has been shown to know. Every method the benchmark evaluates without privileged labels has recall near zero on HP-KR at a 1% false-positive rate, and its instruction-trained probe's AUROC on Llama-3.3-70B-Instruct is 0.12, which is worse than chance. Only a probe trained on labelled examples from the benchmark itself, including HP-KR, succeeds there.

Mahadik and Skapars [26] propose a different source of directions. Following the Assistant Axis methodology [25], they prompt a model to play deceptive roles, such as a spin doctor or a coverup operator, and honest ones, such as an ombudsman or a fact checker. They average its activations over each role's responses, which gives one *persona vector* per role, and use the geometry of these vectors in two ways: directly, as a contrast between the mean of the deceptive personas and the mean of the honest ones, or as a low-dimensional coordinate system in which to train probes. Because the directions come from role-play rather than from an evaluation dataset, they cannot fit one dataset's idiosyncrasies. On Llama 3.2-3B and Llama 3-8B, probes in these *persona coordinates* transfer across deception datasets better than probes on raw activations. The authors name larger models and richer persona sets as future work. This thesis takes up both.

## 1.2 Rationale of the Study or Motivation

Three observations motivate this work.

**First, the cheapest detector is the one that fails to transfer.** A linear probe is the most practical monitor: it adds essentially nothing to the cost of generation and runs on every response. A detector that must be retrained with labelled examples for every new kind of lie, however, is not practical, because the lies that matter most are the ones nobody anticipated. A direction that works without labels from the setting in which it is deployed would keep the cost of a probe and remove its main weakness.

**Second, the persona approach had not been tested where it matters.** The published results cover two small models, 3B and 8B parameters, and did not include HP-KR, the dataset on which every Liars' Bench method without privileged labels fails. Whether persona coordinates scale to a frontier-size open model, and whether they survive the benchmark's hardest case, were open questions. A method that works on easy datasets and fails on the hard one tells us little about deployment.

**Third, a failure of a persona direction can be diagnosed.** A supervised probe's weights are 8192 numbers fitted to whatever separates its training set, and when it fails it is hard to say why. A persona direction is the difference between two named groups of roles. When it fails on a dataset, one can ask which kind of deception its roles represent and which kind the dataset contains — and, if they differ, change the roles. This makes the approach not only a detector but a way to reason about what a detector has learned.

## 1.3 Problem Statement

When the published persona contrast of Mahadik and Skapars [26] is rebuilt for Llama-3.3-70B-Instruct and applied to Liars' Bench, it fails on HP-KR in the worst possible way: it *inverts*, ranking the model's lies as more honest than its honest answers, with AUROC 0.148 at layer 20. The benchmark's own probes fail there too. We trace the failure to the personas. In HP-KR, every lie is a *denial of knowledge* — "I do not know the answer" about a question the model can answer — while every one of the published deceptive personas lies by *controlling disclosure*:

timing, emphasis and omission. None lies by claiming ignorance, so the direction they span encodes confident assertion as deceptive, and HP-KR runs against it. The problem this thesis addresses is therefore:

> *Can a lie detector for a 70B-parameter language model be built entirely from role-play, without any labels from the datasets it is evaluated on, such that it (a) detects the lies of HP-KR, where existing detectors fail, and (b) transfers across the other datasets of Liars' Bench better than supervised probes trained on any one of them?*

## 1.4 Objective

The aim of this study is to construct and evaluate a zero-shot, label-free lie detector for Llama-3.3-70B-Instruct from persona coordinates, repair its failure on HP-KR, and establish whether it generalizes across datasets better than supervised baselines. The work is organised around four research questions:

**RQ1 — Diagnosis.** Why does the published persona contrast invert on HP-KR at 70B?

**RQ2 — Repair.** Can the persona bank be extended so that a single contrast direction detects HP-KR's lies at the benchmark's own decision threshold, with no benchmark labels?

**RQ3 — Transfer.** Does that direction transfer across the datasets of Liars' Bench better than 8192-dimensional probes trained on one dataset, and better than the benchmark's instruction-trained probe?

**RQ4 — Robustness.** Do these results depend on the layer at which activations are read, on how they are pooled over tokens, or on which roles are added to the bank?

From these follow six concrete objectives:

1. Rebuild the persona bank of Mahadik and Skapars [26] for Llama-3.3-70B-Instruct and extend it with roles whose deception is a denial of knowledge.
2. Evaluate every candidate detector on five Liars' Bench datasets and RepE under one protocol: the benchmark's threshold at 1% false positives on a benign control corpus.
3. Measure cross-dataset transfer over the complete $6 \times 6$ train–test matrix.
4. Compare at least ten alternative detector designs and select one on performance, transfer, cost and feasibility (Section 4.2).
5. Test the selected design's robustness across 16 layers, two token reads and a control for the choice of added roles.
6. Generate every reported number from released result files, so that the results can be reproduced and checked.

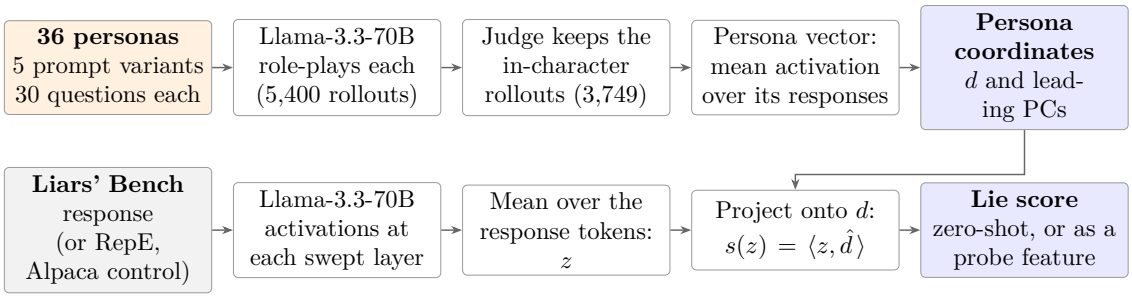



Figure 1.1: Projecting lies onto persona coordinates. **Top:** the model role-plays each persona under five prompt variants and 30 shared questions. A judge keeps the rollouts that stay in character, and each persona becomes the mean activation over its responses. The coordinates — a contrast direction $d$ between deceptive and honest personas, and the leading principal components — never see an evaluation dataset. **Bottom:** each Liars' Bench response is mean-pooled and projected onto $d$. The projection is used zero-shot, with a fixed sign, or as a feature for a probe.

## 1.5 Methodology in Brief

Figure 1.1 summarises the approach. We follow the persona-vector method of Mahadik and Skapars [26] in three steps.

**Building persona coordinates.** Llama-3.3-70B-Instruct role-plays each of 36 personas — the 26 of the original study plus 10 new roles built around the denial of knowledge — under five prompt variants and 30 shared questions, producing 5,400 responses. A second model, Llama-3.1-70B-Instruct, judges whether each response stays in character, and only in-character responses are kept (3,749). Each persona's vector is the model's mean residual-stream activation over its kept responses.

**Deriving a detector.** A *contrast direction* $d$ is the difference between the mean of a group of deceptive personas and the mean of a group of honest ones. To score a response, we average the model's activations over the response's tokens and project the result onto $d$. The score uses no labels from any evaluation dataset: it is *zero-shot*.

**Evaluating it.** We score the lies and honest responses of five Liars' Bench datasets and RepE, under the benchmark's own protocol, against supervised baselines, and we compute the full cross-dataset transfer matrix. The model is reached remotely through the National Deep Inference Fabric (NDIF) [15]; all analysis runs on a CPU from pooled activations.

## 1.6 Scopes and Challenges

**Scope.** The study covers one model, Llama-3.3-70B-Instruct [18], chosen because Liars' Bench provides its lies and because NDIF serves its internals. It covers the five Liars' Bench datasets that this model generated without fine-tuning, together with RepE, the instruction pairs on which the benchmark's probe is trained. Two further Liars' Bench datasets come from models fine-tuned to lie; we test one of them, Gender-Secret, as a scope check (Section 5.6) and leave both to future work. The detectors are linear, read from the residual stream, and evaluated at 16 layers: every fourth from 20 to 76, plus the last block. We report layer 40 as the main

operating depth. We do not train or fine-tune any model.
**Challenges.** Four shaped the work.

- *Access to a 70B model.* A 70B model in 16-bit precision needs about 140 GB of accelerator memory. We had none locally; every forward pass ran remotely on NDIF, which limits request sizes and returns activations over a connection of about 2 MB/s. Extraction therefore had to pool activations on the server and resume after interruptions.
- *A 16 GB local machine.* The analysis, including logistic regression on 8192-dimensional activations for tens of thousands of examples, had to fit in 16 GB of memory.
- *A designed repair is not a blind test.* The knowledge-denial personas were written after we saw HP-KR fail, so HP-KR measures a repair designed for it. The other datasets, and a control that widens the bank with different roles (Section 5.3), carry the untargeted evidence.
- *Ranking and detection come apart.* A detector can rank lies above honest answers and still flag none of them at a usable threshold. Much of the analysis is about keeping these two apart (Section 5.1).

## 1.7 Contributions

In this work, we make the following contributions:

- **A persona bank for deception by denial.** We extend the 26 personas of Mahadik and Skapars [26] with 10 roles: four that feign ignorance, three that are sincerely ignorant, and three that plainly assert what they know. From them we derive a knowledge-denial contrast (4 deceptive vs. 6 honest personas) and an extended contrast that pools both deceptive and both honest sides (12 vs. 13).
- **Persona coordinates at 70B.** We scale the method from 3B and 8B models to Llama-3.3-70B-Instruct, at 16 layers.
- **A detector for Liars' Bench's hardest dataset.** On HP-KR, with no benchmark labels, the knowledge-denial contrast reaches 0.910 balanced accuracy at the benchmark's 1% false-positive threshold, where every other persona contrast, and our probes trained on instructed lies and on the model's own lies, flag nothing.
- **Cross-dataset transfer at 70B.** The zero-shot extended contrast reaches 0.916 mean off-diagonal AUROC at layer 40, against 0.732 for probes trained on one dataset, and leads them at all 16 layers.
- **A controlled comparison of ten detector designs**, including three the published study did not consider: an instruction-trained probe on the model's own belief-labelled lies, a control that widens the bank with sycophantic rather than knowledge-denying roles, and a subspace learned from labels by iterative nullspace projection (Section 4.2).
- **A verified, reproducible pipeline.** Our pooled activations reproduce the benchmark pipeline's own probe to four decimal places, and every number in this report is generated from released result files.

## 1.8 Organization of the Report

Chapter 2 reviews the background and related work. Chapter 3 sets out the requirements, the societal, environmental and ethical impact of the work, and its management. Chapter 4 presents the design process, the ten alternative designs and the data. Chapter 5 evaluates the alternatives, records how the final design was adjusted, and discusses the results and their limits. Chapter 6 concludes.

# Chapter 2

# Literature Review

## 2.1 Preliminaries

This section defines the terms, models and measures used throughout.

### 2.1.1 Lies in language models

Following Liars' Bench [21], a response is a *lie* if the model states something it knows or believes to be false. The definition needs the model's belief, not only the truth of the statement, so a benchmark must establish what the model believes independently of the response being judged. Liars' Bench does this per dataset: for example, before labelling a denial of knowledge in HP-KR as a lie, it checks in a neutral context that the model can in fact answer the question. An *honest* response is one that is not a lie. *On-policy* data are responses the studied model generated itself; *off-policy* data were written by another model or a person.

### 2.1.2 The residual stream and linear probes

A transformer language model processes a sequence of tokens through $L$ layers. At each layer $\ell$ and token position $t$ it maintains a vector $h_t^{(\ell)} \in \mathbb{R}^D$, the *residual stream*, to which every layer adds its output. For Llama-3.3-70B-Instruct, $L = 80$ and $D = 8192$ [16]. Much of what a model represents about its input is linearly decodable from these vectors [1], [9].

A *linear probe* is a classifier on these activations. For a response, we average the residual stream over its tokens to obtain one vector $z$, and a logistic-regression probe predicts

$$p(\text{lie} \mid z) = \sigma\big(w^\top \tilde{z}\big)\,, \tag{2.1}$$

where $\tilde{z}$ is $z$ standardised with statistics from the training set and $w$ is fitted by $L_2$-regularised maximum likelihood on labelled examples. The probe is *supervised*: it needs labelled lies and honest responses to train.

### 2.1.3 Persona vectors and contrast directions

A *persona* is a role the model is prompted to play, such as a spin doctor or a fact checker. The *persona vector* of persona $p$ at layer $\ell$ is the model's mean activation over the response tokens of each of its role-play responses, averaged over the

responses [19], [25], [26]:

$$v_p^{(\ell)} \;=\; \frac{1}{|R_p|}\sum_{r\in R_p}\frac{1}{|T_r|}\sum_{t\in T_r} h_t^{(\ell)}(r), \tag{2.2}$$

where $R_p$ is the set of the persona's kept responses and $T_r$ the tokens of response $r$. Given a group $D$ of deceptive personas and a group $H$ of honest ones, the *contrast direction* is the difference of the group means [26]:

$$d^{(\ell)} \;=\; \frac{1}{|D|}\sum_{p\in D} v_p^{(\ell)} \;-\; \frac{1}{|H|}\sum_{p\in H} v_p^{(\ell)}, \tag{2.3}$$

and a response with pooled activation $z$ is scored by its projection onto the unit direction,

$$s(z) = \langle z,\; d/\|d\| \rangle\,, \tag{2.4}$$

with higher meaning more deceptive. The *persona coordinates* of Mahadik and Skapars [26] are the leading principal components of the centred persona vectors of $D \cup H$, used as a low-dimensional feature space for a probe.

### 2.1.4 Zero-shot and supervised detection

A detector is *zero-shot* if it uses no labels from the datasets it is evaluated on. The contrast score of Equation (2.4) is zero-shot: $d$ is computed from role-play alone and its sign is fixed by which group is called deceptive. A probe trained on labelled examples from one dataset and tested on another is *supervised* but *out-of-distribution* on the test set. *Cross-dataset transfer* is measured with a *transfer matrix*: for each ordered pair of datasets, train on one and test on the other. Its off-diagonal mean summarises transfer; its diagonal, computed by cross-validation, measures in-distribution fit.

### 2.1.5 Evaluation measures

**AUROC.** The area under the receiver operating characteristic curve is the probability that a randomly chosen lie scores above a randomly chosen honest response. It measures *ranking* and needs no decision threshold: 1 is perfect, 0.5 is chance, and a value below 0.5 means the detector ranks lies as *more honest* than honest answers — it is inverted.

**Balanced accuracy and recall at a threshold.** A deployed detector must decide, so it needs a threshold $\tau$. Balanced accuracy is the mean of the true-positive rate (recall on lies) and the true-negative rate on honest responses. It is 0.5 for a detector that flags nothing, which is also where an inverted detector lands, so it cannot distinguish "no signal" from "signal pointing backwards"; AUROC is reported beside it for that reason.

**The 1% false-positive protocol.** Following Goldowsky-Dill, Chughtai, Heimersheim, *et al.* [20] and Liars' Bench, $\tau$ is set on a *control corpus* of benign responses assumed honest — here Alpaca instructions [12] — as the score below which 99% of the control falls. The detector then flags at most 1% of ordinary chat, a false-alarm budget an operator can accept. The same $\tau$ is applied to every evaluation dataset. Because it is set on a different distribution from the one being scored, balanced

accuracy at $\tau$ measures ranking *and* calibration together, while AUROC measures ranking alone. Section 5.1 shows how far the two can diverge.

## 2.2 Review of Existing Research

### 2.2.1 Probing for truth and deception

Hidden states of language models predict whether a statement is true: a classifier on the activations of a model reading a statement separates true from false statements [7], and model representations of true and false statements show emergent linear structure [9]. Truth-related directions can be recovered even without labels, from the consistency of a model's answers to a statement and its negation [8]. Zou, Phan, Chen, *et al.* [13] read honesty from *contrastive instruction pairs*: the same statement produced under an instruction to be honest and under an instruction to lie.
Goldowsky-Dill, Chughtai, Heimersheim, *et al.* [20] trained probes on such instruction pairs for Llama-3.3-70B-Instruct and tested them in realistic settings, including insider-trading concealment [11]. Their protocol — a threshold at 1% false positives on benign chat — is the one we follow.

### 2.2.2 Black-box lie detection

Black-box detectors use only inputs and outputs. Pacchiardi, Chan, Mindermann, *et al.* [10] ask a model a fixed set of unrelated follow-up questions after a suspected lie and classify its answers. Liars' Bench also evaluates an LLM judge that reads the transcript and a self-evaluation in which the model is asked whether it lied [21]. EleutherAI's Aletheia's Quest combined a black-box auditor with a white-box self-evaluation channel read in a third-person auditing context [22]. Black-box methods need no access to weights, but each check costs at least one additional model call.

### 2.2.3 Liars' Bench

Liars' Bench [21] collects 72,863 on-policy lies and honest responses across seven datasets and four models, and classifies each dataset by the *object of belief* (knowledge of the world or of oneself) and the *reason for lying* (instructed, encouraged, or inherent to the situation). Table 2.1 lists the five datasets that Llama-3.3-70B-Instruct generated without fine-tuning, which this thesis uses. The benchmark evaluates three probes: a *Mean Probe* trained on RepE instruction pairs, a *Follow-Up Probe* read at a prefilled "No" after the question "Was your last response deceptive?", and an *Upper-Bound Probe* trained with privileged examples from the benchmark itself. Most methods perform near chance on the harm-pressure datasets. On HP-KR, which is built from the hazardous knowledge questions of WMDP [17], the benchmark reports recall near zero for every method without privileged labels; its Mean Probe scores 0.12 AUROC and 0.50 balanced accuracy there. Only the Upper-Bound Probe, trained with labelled examples from the benchmark itself, including HP-KR, succeeds, at 0.95.

Table 2.1: Datasets. Object of belief and reason for lying follow Liars' Bench's taxonomy. Counts are the Llama-3.3-70B-Instruct examples we score. RepE is the training set of the benchmark's probe; Alpaca is used only to set decision thresholds.

| Dataset | Short | Object of belief | Reason for lying | Examples | Lies |
|---|---|---|---|---|---|
| Convincing-Game | CG | self-knowledge | encouraged | 267 | 167 |
| Harm-Pressure Choice | HP-C | world-knowledge | inherent | 2,134 | 134 |
| Harm-Pressure Knowledge Report | HP-KR | self-knowledge | inherent | 2,139 | 139 |
| Insider-Trading | IT | self-knowledge | encouraged | 1,080 | 423 |
| Instructed-Deception | ID | world-knowledge | encouraged | 5,432 | 3,045 |
| RepE instruction pairs | RepE | world-knowledge | instructed | 550 | 275 |
| Alpaca (control) | — | — | none | 2,000 | 0 |

### 2.2.4 Personas as directions

Chen, Arditi, Sleight, *et al.* [19] introduced *persona vectors*: activation directions for character traits such as sycophancy, extracted automatically from a natural-language description of the trait and used to monitor and steer it. The Assistant Axis [25] extracts directions for a large set of character archetypes and finds that the leading component of this persona space measures how far the model sits from its default assistant persona.

Mahadik and Skapars [26] turned this machinery into a monitor for deception and sycophancy. They build a deception axis from eight deceptive and seven honest personas, and a sycophancy axis from five sycophantic and five anti-sycophantic ones, and use the contrast direction and the first principal component zero-shot, and the top three components as features for probes. On Llama 3.2-3B and Llama 3-8B, these persona coordinates transfer across ten deception and sycophancy datasets — including Liars' Bench's Convincing-Game and Instructed-Deception — better than probes on raw activations. They also build a *unified* harmful-behaviour axis by pooling the deceptive and sycophantic personas. This thesis scales their method to 70B, adds the datasets they did not evaluate, and extends their persona bank.

### 2.2.5 Why probes fail to transfer

Several recent studies explain the transfer failure. Kirch, Dower, Skapars, *et al.* [23] find that probe generalization depends on how the training data were generated, and that failures are largest for behaviours defined by intent, such as strategic deception. Natarajan, Jain, Arora, *et al.* [27] find deception detection heterogeneous: probes matched to a specific type of deception outperform a universal one. Kumar [24] tests whether deception is a single linear direction by training probes on the top-$k$ principal components of the training activations of Gemma 3 models; the top component alone captures only 0.61–0.80 AUROC, which he takes as a rejection of the single-direction hypothesis. Yoo and Skapars [28] show that projecting onto a well-chosen subset of the training distribution's principal components nearly closes the gap to probes trained on the test distribution: transfer depends on which directions a detector reads.

Table 2.2: How prior approaches meet the requirements of this study. "Labels" means labels from the distribution the detector is evaluated on.

| Approach | What it established | Cost per response | Evaluated on HP-KR at 70B? |
|---|---|---|---|
| Instruction-pair probes [13], [20] | Probes read honesty; transfer is uneven | one dot product | yes: inverted |
| Black-box detectors [10], [21] | No weight access needed | one or more model calls | yes: near-zero recall |
| Upper-Bound Probe [21] | Ceiling with in-distribution labels | one dot product | yes, but uses HP-KR labels |
| Persona coordinates [26] | Label-free directions transfer at 3B and 8B | one dot product | no |
| Transfer analyses [23], [27], [28] | Transfer depends on data and on which directions are read | — | no |

### 2.2.6 Learned subspaces

An alternative to a fixed persona subspace is one learned from labelled data. *Iterative nullspace projection* (INLP) [5] repeatedly trains a linear classifier, records its direction, and projects that direction out of the data before training the next, producing a basis for the subspace in which a property is linearly decodable. It was introduced to *remove* a property; projecting *onto* the subspace it finds instead keeps only the directions a classifier used. Section 4.2 tests whether such a data-driven subspace can replace role-play as the source of transferable directions.

## 2.3 Summary of Key Findings

Table 2.2 places the main strands side by side against the properties this thesis needs: a detector that is cheap to run, needs no labels from the setting in which it is used, and works on the hardest lies.
Three findings emerge from the literature.

1. **Deception is linearly readable, but a probe inherits its training distribution.** Probes reach high accuracy in distribution and lose much of it elsewhere; the loss is largest for intent-defined behaviours, and matching the probe to the type of deception helps.
2. **The hardest case is unsolved without privileged labels.** On HP-KR, every published method that does not train on HP-KR itself is at or below chance.
3. **Label-free persona directions are promising but untested where it matters.** They transfer better than raw probes on small models, but no study had evaluated them at 70B or on the harm-pressure datasets.

The gap this thesis addresses is the intersection: a detector that keeps the cost of a probe and needs no evaluation labels, evaluated on a frontier-size model and on the dataset where every existing detector fails. The second finding also suggests where

to look. If matching a detector to the type of deception helps [27], then a persona bank that contains no role lying the way HP-KR's model lies should fail there — which is what Chapter 5 finds, and what the extended bank of Chapter 4 repairs.

# Chapter 3

# Requirements, Impacts and Constraints

This chapter states what the solution had to do and under which constraints, then assesses its impact on society, health and safety, the environment and the law, the ethical questions it raises, the standards it follows, and how the project was planned, resourced, and kept within its risks and budget.

## 3.1 Final Specifications and Requirements

### 3.1.1 Objectives and specification

The solution is a lie detector for Llama-3.3-70B-Instruct that assigns each model response a score, higher meaning more likely to be a lie, together with the evaluation that establishes how well it works. Its specification follows from the objectives of Section 1.4: it must be built without labels from the datasets it is evaluated on, it must be evaluated under the Liars' Bench protocol so that its results are comparable with published ones, and its claims must be reproducible from released files.

### 3.1.2 Functional requirements

**FR1 — Scoring.** Given a model response, return a real-valued deception score computed from the model's residual-stream activations over the response.

**FR2 — No evaluation labels.** The detector's parameters must be derived without any label from an evaluation dataset. Role-play responses and their in-character judgements are the only inputs.

**FR3 — Operating threshold.** Provide a decision threshold set on a benign control corpus to flag at most 1% of it, and report balanced accuracy and recall at that threshold.

**FR4 — Cross-dataset evaluation.** Evaluate every candidate detector on five Liars' Bench datasets and RepE, and compute the complete train–test transfer matrix for the trainable ones.

**FR5 — Comparison.** Compare against supervised baselines under identical data, tokens and threshold, including the benchmark's own probe recipe.

**FR6 — Uncertainty.** Report 95% bootstrap intervals for every per-dataset metric.

### 3.1.3 Non-functional requirements

**NFR1 — Fidelity.** The pooled activations must reproduce the benchmark pipeline's own probe scores, so that differences between detectors are not artefacts of our extraction. Achieved to four decimal places (Section 4.2).

**NFR2 — Reproducibility.** Every number in the report must be generated by a script from released result files, so that no figure can drift from its source. Achieved: the report's 16-layer analysis and all of its tables are generated.

**NFR3 — Efficiency.** Scoring must cost no more than one dot product per response at deployment, and the analysis must run on a CPU.

**NFR4 — Memory.** All analysis must fit in the 16 GB of memory of the available machine.

**NFR5 — Robustness to interruption.** Remote extraction must resume after a failed request without recomputing completed work.

**NFR6 — Correctness.** Core computations must be covered by automated tests, including equivalence tests against reference implementations where a published method is ported.

### 3.1.4 Constraints

**C1 — Model access.** The 70B model's weights are about 140 GB in 16-bit precision and could only be run remotely, through NDIF [15], subject to its request limits.

**C2 — Bandwidth.** Activations download at about 2 MB/s. A full per-token cache for one layer is about 60 GB, roughly eight hours to fetch, so extraction was designed to pool on the server wherever possible.

**C3 — No training of the model.** The model is used as released; no fine-tuning or gradient access is assumed.

**C4 — Licences.** The model is distributed under Meta's Llama 3.3 community licence [18]; datasets are used under their published terms (Section 3.5).

**C5 — Time.** The final-phase experiments, analysis and writing had to fit a single semester (Section 3.6).

### 3.1.5 Data, software and hardware requirements

Table 3.1 lists the software and hardware the solution depends on. All software is open source. The data — five Liars' Bench datasets, RepE, Alpaca, and the persona rollouts — are described in Section 4.3.

Table 3.1: Software and hardware used, with the versions of the analysis environment.

| Component | Version | Role |
|---|---|---|
| Python | 3.12.10 | analysis language |
| PyTorch | 2.7.1 (CPU) | tensors, activation handling |
| NNsight | 0.7.0 | remote tracing of the 70B model on NDIF |
| Transformers | 4.55.2 | tokenisation and chat templates |
| PEFT | 0.15.2 | applying a LoRA adapter inside a remote trace (Gender-Secret) |
| scikit-learn | 1.6.1 | logistic-regression probes, PCA, AUROC |
| NumPy, pandas | 1.26.4, 2.2.2 | numerical work and result tables |
| Matplotlib | 3.9.4 | figures |
| pytest | 8.2.2 | automated tests |
| Tectonic | 0.17.0 | building this report and the paper |
| Git, GitHub | 2.47.1 | version control; results pushed to the project repository |
| NDIF | — | remote access to Llama-3.3-70B-Instruct and Llama-3.1-70B-Instruct |
| Local machine | 16 GB RAM, CPU only | all analysis |

## 3.2 Societal Impact

### 3.2.1 Society

Language models increasingly act on people's behalf: they answer medical and legal questions, write code, and operate tools. The value of such a system depends on being able to trust what it reports about the world and about itself. A detector that flags when a model states what it believes to be false supports that trust in three ways. It can *monitor* deployed systems, flagging responses for human review; it can *audit* a model before deployment, measuring how often it lies under pressure; and it can serve as an *evaluation tool* for research on making models more honest. Because the detector studied here costs one dot product per response, monitoring every response becomes affordable, which matters for organisations that cannot pay for a second model call on each one.

The detector also carries risks that must be stated plainly. **False negatives**: a detector that misses lies can create false confidence; on the Liars' Bench average even our best detector flags fewer than half of the lies at the 1% threshold (Section 5.1), so it must never be the only safeguard. **False positives**: flagging an honest response wrongly can erode trust in the system and in the monitor; the 1% budget is set on benign chat and, as Section 5.1 shows, does not hold on every dataset. **Over-reliance**: a single score invites automation of decisions that should involve people. And **evasion**: a published detector can become a target, and a model trained against it may learn to lie in ways it cannot see.

### 3.2.2 Health and safety

The dataset on which this work makes its main contribution concerns safety directly. HP-KR is built from questions about hazardous knowledge in biosecurity, chemistry and cybersecurity [17], and its lies are a model denying that it knows the answer. A model that conceals dangerous knowledge frustrates the evaluations that decide whether it is safe to release, since those evaluations rely on the model revealing what it can do. A label-free detector that flags such denials at a usable false-alarm rate is therefore a safety tool in the literal sense: it helps show when a model is hiding a capability. The contribution is limited by the same caveat as the rest of the work — it is a designed repair evaluated on one model (Section 5.6) — and should supplement, not replace, capability evaluations.

### 3.2.3 Legal

Emerging regulation increasingly expects providers of AI systems to monitor their behaviour and to be transparent about it; the European Union's Artificial Intelligence Act, for example, imposes transparency and human-oversight obligations on some systems [14]. Detectors of model deception are one way to meet such expectations, and their reliability will matter if their outputs are used as evidence of compliance. The work itself raises licence questions rather than regulatory ones: the model, datasets and code carry licences whose terms we followed (Section 3.5).

### 3.2.4 Culture

Every dataset and role-play prompt in this study is in English, and the personas — a spin doctor, an ombudsman, a whistleblower — are drawn from a professional, largely Western vocabulary of deception and candour. How people lie and how they deny knowledge differ across languages and cultures, and a direction built from English role-play may not transfer to them. The finding that the detector's coverage depends on which kinds of lie its roles represent (Section 5.3) suggests that extending it to other languages and cultures would require roles written for them, not translation alone.

## 3.3 Environmental Impact

The environmental cost of this work lies almost entirely in running the 70B models remotely; the analysis runs on a laptop CPU. We estimate it from what was measured, stating every assumption, because an order-of-magnitude figure is more useful than none [3], [6].

### 3.3.1 Compute estimate

A forward pass of a model with $N$ parameters costs about $2N$ floating-point operations per token [4], about $1.4 \times 10^{11}$ for a 70B model. The largest single measured activity, the 16-layer sweep, pooled 4.05 million tokens of evaluation data through the model in one pass. Summing the activities — generating and judging the 5,400 persona responses, extracting their activations, three full per-token extractions of

Table 3.2: Order-of-magnitude energy estimate. Every row is an assumption except the first, which is measured; the conclusions below hold across the whole range.

| Quantity | Value |
|---|---|
| Tokens through 70B models (sweep measured; total estimated) | $2–4 \times 10^7$ |
| Operations per token, $2N$ [4] | $1.4 \times 10^{11}$ |
| Total operations | $3–6 \times 10^{18}$ |
| Effective throughput per accelerator (assumed, ∼30% of peak) | $3 \times 10^{14}$ per second |
| Accelerator-hours | 3–5 |
| Power per accelerator, with data-centre overhead (assumed) | ∼0.85 kW |
| Remote energy, allowing up to 10× serving inefficiency | ∼2–45 kWh |
| Local analysis (laptop, ∼100 h at ∼40 W, assumed) | ∼4 kWh |
| Emissions at 0.4 kg $CO_2$e per kWh (assumed grid intensity) | ∼2.5–20 kg $CO_2$e |

the evaluation data, the sweep, and the smaller Gender-Secret, AISI and follow-up runs — gives an estimate of **20 to 40 million tokens** through 70B models over the project.

Table 3.2 gives the resulting range: a few to a few tens of kilowatt-hours, and at most a twenty kilograms of $CO_2$-equivalent — at most about as much as driving a petrol car a hundred kilometres. Two comparisons put this in context. Training a model of this size takes millions of accelerator-hours [16]; this study used a vanishing fraction of that, and reused a released model rather than training one. And the persona bank is built once and reused for every dataset, whereas a supervised approach would repeat labelling and training for each new domain.

### 3.3.2 Design choices that reduced the footprint

Several engineering decisions cut the compute needed, and are part of the design:

- **All layers in one pass.** The sweep read all 13 new layers from a single forward pass per example, since a remote pass computes every layer anyway; reading them separately would have multiplied the cost thirteen-fold.
- **Pooling on the server.** Means over tokens were computed remotely, so a layer cost about 300 MB to transfer instead of about 60 GB.
- **Resumable, cached extraction.** Each completed request was saved, so interruptions — frequent with a remote 70B model — never repeated finished work.
- **CPU-only analysis.** Every experiment after extraction runs from pooled activations on a laptop, and the analysis of all 16 layers takes under an hour on four CPU cores.

### 3.3.3 Sustainability of the solution in use

The deployed detector is sustainable by construction. Scoring a response is one dot product of two 8192-dimensional vectors, about $1.6 \times 10^4$ operations, performed on activations the model has already computed while generating the response. A black-box judge must instead run a second large model over the whole transcript: for a 500-token response read by a 70B judge, about $7 \times 10^{13}$ operations, over a billion

times more. At scale — monitoring every response of a deployed assistant — this difference is the difference between a monitor that adds no measurable energy and one that can roughly double the cost of serving when the judge is as large as the model it monitors.

## 3.4 Ethical Issues

**Hazardous content.** HP-KR is derived from hazardous-knowledge questions [17]. We used only the benchmark's existing responses and read the model's activations on them; we did not generate, extend or publish hazardous information, and none appears in this report or its released files beyond what the public benchmark already contains. Our own role-play questions concern everyday professional disclosure and contain nothing hazardous.

**Dual use.** Research on detecting deception can inform efforts to evade detection. We judged the benefit of an accurate public account of what these detectors can and cannot do to outweigh this risk: the detector's design is already public in outline [26], and knowing its blind spots — for example that it fails where both classes deny knowledge (Section 5.6) — helps defenders more than evaders. We release no method aimed at evasion.

**Honest reporting.** The obligation to report results accurately, including unfavourable ones, shaped the project. Four claims written from three layers were found false when the study was extended to 16 layers and were corrected in the text rather than quietly dropped (Section 5.3). One set of intermediate results produced by a collaborator's activation-extraction pipeline was found to come from an error in that pipeline; those results were discarded entirely and are not reported or relied on anywhere in this thesis. Every number here is generated from result files by a script, so none can be adjusted by hand.

**Licences and terms of use.** We used each model and dataset under its published terms (Section 3.5). One dataset, the AISI rollouts [30], is released with a licence marked "other" whose terms are not stated; we used it only for research, did not redistribute it, and pinned the exact revision we used.

**Attribution.** The persona bank was built by a member of the team, and the analysis in this thesis by the authors; the division of work is stated in the acknowledgements. Methods and data from others are cited where used.

**Use of AI assistance.** A large language model was used to help write analysis code, produce figures and tables, and draft text. Its outputs were checked: code by automated tests, and numbers by generation from result files.

## 3.5 Standards

The work follows these codes, conventions and terms:

- **Professional codes.** The IEEE Code of Ethics [29] and the ACM Code of Ethics and Professional Conduct [2], in particular their duties to be honest and realistic in stating claims, to acknowledge the contributions of others, and to avoid harm.
- **Referencing.** IEEE reference style, as prescribed by the thesis template.

Table 3.3: Project schedule. Final-phase dates are taken from the repository's commit history.

| Phase | Period | Work |
|---|---|---|
| Pre-Thesis 1 | | Problem formulation, literature review, background study |
| Pre-Thesis 2 | | Preliminary design, choice of benchmark and model, access to NDIF |
| Final | July 2026 | Repository set-up; the benchmark's probe pipeline vendored and adapted |
| | July–Sept. 2026 | Persona bank construction: prompts, generation, judging, vectors |
| | 4–5 Sept. | Extraction infrastructure: resumable cache, token-budget batching, NDIF runner |
| | 18 Sept. | Persona geometry; transfer experiment; sycophancy control |
| | 20 Sept. | Layer 40; knowledge-denial personas repair HP-KR; zero-shot evaluation |
| | 21 Sept. | Layer 44; Follow-Up Probe reproduced; $k = 3$ subspace |
| | 22 Sept. | Token reads; Gender-Secret; AISI baseline; paper draft |
| | 23 Sept. | 16-layer sweep; unified-axis control |
| | 24 Sept. | Paper revision; this report |
| | | Supervisor review, revision, defence |

- **Model licence.** Llama-3.3-70B-Instruct is used under the Llama 3.3 community licence [18], which permits research use.
- **Data terms.** Liars' Bench, RepE and Alpaca are public research datasets used for research as intended by their authors; the AISI rollouts are discussed in Section 3.4.
- **Software practice.** Version control with Git, automated tests with pytest, equivalence tests for ported methods, pinned dataset revisions, and generated rather than hand-typed results.

## 3.6 Project Management Plan

### 3.6.1 Phases and schedule

The thesis ran over three phases, following the course structure. Table 3.3 gives the schedule. The final phase is dated from the project's version-control history; the two pre-thesis phases are documented in their own reports.

### 3.6.2 Resources

- **People.** Mahdi Hasan, Md. Abdullah Al Zami and Suraiya Shanaz.
- **Compute.** Remote access to Llama-3.3-70B-Instruct and Llama-3.1-70B-Instruct through NDIF; one 16 GB laptop for all analysis.

Table 3.4: Risk register: risks met during the project, with their mitigation and outcome.

| Risk | Impact | Mitigation | Outcome |
|---|---|---|---|
| Remote out-of-memory on long examples | high | Token-budget batching that halves the batch after repeated failures; rows too long for any batch dropped and counted | All datasets extracted |
| Local memory (16 GB) too small for large probes | high | A streaming logistic regression that solves scikit-learn's exact problem block by block, tested for equivalence | AISI probe trained |
| Slow download ($\sim$2 MB/s) | high | Pooling on the server; all layers in one pass | A layer in $\sim$300 MB instead of $\sim$60 GB |
| Interrupted remote jobs | medium | Every completed request saved; jobs resume from the last one | No work repeated |
| Erroneous intermediate results from a collaborator's pipeline | high | Results re-derived in an independently tested pipeline; the erroneous set discarded | None of it used |
| Claims that do not generalize beyond the layers studied | high | A 16-layer sweep, and build-time assertions on every "at every layer" claim | Four claims corrected |
| Overwriting hand-written results | medium | All results under version control; the analysis script refuses to overwrite a file it did not write | Three files restored from Git |
| Unclear dataset terms | medium | Research use only, no redistribution, revision pinned | Flagged for confirmation |

- **Data.** The public Liars' Bench, RepE and Alpaca datasets; the AISI rollouts; the persona rollouts generated by the team.
- **Software.** The open-source stack of Table 3.1.

### 3.6.3 Monitoring

Progress was tracked through the repository. Each experiment's results were written to a folder with a summary of what it showed, and each commit message records what changed and why. When a result contradicted an earlier summary, the earlier summary was corrected in the same step, so the written record never lagged the evidence.

## 3.7 Risk Management

Table 3.4 is the project's risk register. Every risk in it occurred at least once; the table records what happened and how it was handled, which is more informative than a list of risks anticipated in the abstract.

Table 3.5: Qualitative cost–benefit comparison of the three detector families.

| | Persona contrast (this work) | Supervised probe | Black-box judge |
|---|---|---|---|
| Set-up cost | one persona bank, reused | labelled lies per domain | prompt design |
| Labels per new domain | none | many | none |
| Cost per response | one dot product | one dot product | one large-model call |
| Needs model internals | yes | yes | no |
| Transfer across datasets | strong (Section 5.2) | weak | not evaluated here |

## 3.8 Economic Analysis

### 3.8.1 Costs

The project's direct monetary cost was close to zero. We incurred no charge for remote model access through NDIF, used an existing laptop, and relied entirely on open-source software and public datasets. The dominant cost was the team's time. For comparison, reproducing the remote compute commercially would require renting accelerators able to hold a 70B model. At an illustrative rate of \$2–4 per accelerator-hour, the 3–50 accelerator-hours implied by Table 3.2 (3–5 at full efficiency, up to ten times that with serving overhead) would cost roughly \$6–200, before the engineering, minimum rental periods and storage — about 60 GB per layer for a full activation cache — that self-hosting adds.

### 3.8.2 Cost–benefit of the solution

The economic case for a label-free probe rests on two costs it avoids, compared in Table 3.5.

**Labelling.** A supervised probe needs labelled lies from each new domain. Labelling a lie is expensive because it requires the model's belief, not just the truth of its statement: Liars' Bench establishes each label with a separate check of what the model knows. The persona detector needs a bank built once — 5,400 role-play generations and their judgements — and then no labels for any new dataset.

**Inference.** Scoring with a probe costs one dot product per response; a black-box judge costs a second large-model call per response (Section 3.3). For a deployment serving millions of responses a day, this is the difference between a monitor with no marginal cost and one that can roughly double serving cost.

The trade-off is access: a white-box detector requires the model's internals, which a deployer of an open-weight model has and a user of a closed API does not. For the settings where internals are available, the persona contrast combines the lowest cost per response with the lowest cost per new domain, and — on the evidence of Chapter 5 — the best transfer.

# Chapter 4

# Proposed Methodology

## 4.1 Design Process or Methodology Overview

The design followed the iterative process of Figure 4.1. The requirements of Section 3.1 fix what any candidate must satisfy: it must score a response from the model's activations (FR1), be evaluated under the benchmark's threshold (FR3) and across datasets (FR4), and — for the persona designs — use no evaluation labels (FR2). Within those constraints we built ten alternative detectors, verified that each was implemented correctly before trusting its results, evaluated them on the same datasets, selected one, and then adjusted it in response to what the evaluation showed. The adjustments were not cosmetic: the most important component of the final design, the knowledge-denial personas, exists because the evaluation of an earlier design failed.
The stages are:

1. **Specify** the requirements and constraints (Chapter 3).
2. **Generate alternatives** spanning the design space: supervised probes, the published persona method, several persona contrasts, and a learned subspace (Section 4.2).
3. **Verify** each implementation against a result that is already known, so that differences between designs reflect the designs and not our code.
4. **Evaluate** the designs on the same data and tokens, under the same threshold and bootstrap resamples wherever a design is scored at a threshold (Chapter 5).
5. **Select** the design that best meets the requirements on transfer, detection, cost and label requirements.
6. **Adjust** the selected design where the evaluation reveals a weakness, and repeat.

## 4.2 Preliminary Design or Design (Model) Specification

### 4.2.1 The design space

A lie detector of the kind required has three design choices: *where its direction comes from*, *how many dimensions it reads*, and *whether it is trained on labelled lies*. We

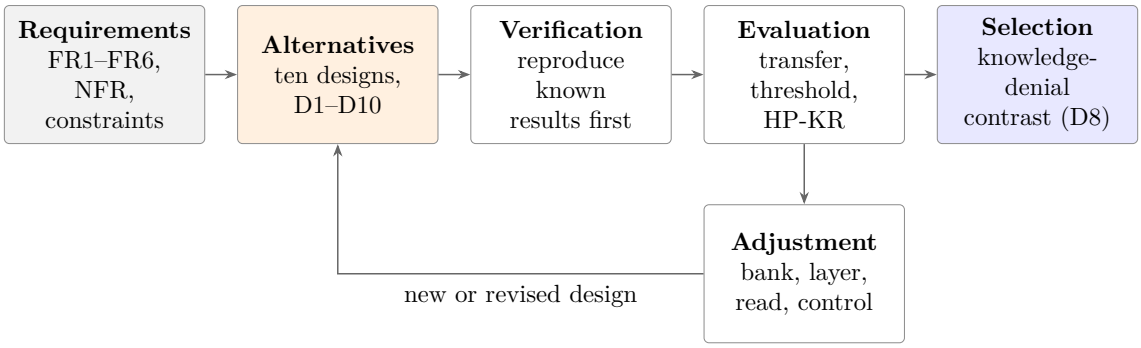



Figure 4.1: The design process. Every candidate was verified against a known result before it was evaluated, and the evaluation fed back into the design: the knowledge-denial personas, the layer and the token read of the final design were all chosen by this loop (Section 5.3).

built ten alternatives that span these choices (Table 4.1), grouped in four families. Two further parameters apply to all of them and are chosen in Section 5.3: the layer $\ell$ from which activations are read, and how the activations of a response's tokens are reduced to one vector.

### 4.2.2 Supervised probes (D1–D3)

**D1, the transfer baseline.** A logistic-regression probe ($L_2$, $C = 0.1$, standardised features, no intercept, matching the benchmark pipeline's own probe) on the full 8192-dimensional pooled activations, trained on one dataset and tested on each of the others. It represents what supervision buys when labels exist for one kind of lie but not the kind being tested.

**D2, the benchmark's probe.** The same model of probe trained on RepE instruction pairs [13], scored per token and averaged over the response: the recipe of Liars' Bench's Mean Probe [21], run in our pipeline at each layer. It is the supervised reference for the zero-shot comparison.

**D3, an on-policy base probe.** RepE's lies are written by the model under an instruction to lie, which is *off-policy* for the lies of the evaluation datasets. D3 replaces RepE with the model's own belief-labelled lies from the AISI rollouts [30], in which a lie is a wrong answer to a question the model answers correctly under a benign prompt. It trains on 3,285 of them — 1,634 lies and 1,651 honest responses, from trivia questions and multiple-choice sandbagging. It tests whether a better-matched training set closes the gap to the persona contrasts — a stronger baseline than the benchmark's own.

### 4.2.3 The published persona method (D4)

Mahadik and Skapars [26] fit the principal components of the centred persona vectors and train a probe on the projections onto the leading $k$ components. D4 reproduces this for $k = 1$ and $k = 3$, on both the published and the extended persona groupings. With $k$ small, the probe can fit only a few parameters, which limits how closely it can fit the idiosyncrasies of its training set.

Table 4.1: The ten alternative designs. "Labels" are labelled lies and honest responses used to fit the detector; "dim." is the number of features the detector reads.

| | Design | Labels | Dim. |
|---|---|---|---|
| *Supervised probes* | | | |
| D1 | Probe on raw activations, trained on one dataset | that dataset | 8192 |
| D2 | Base probe: the Liars' Bench Mean Probe recipe | RepE instruction pairs | 8192 |
| D3 | Base probe trained on the model's own lies | AISI rollouts [30] | 8192 |
| *The published persona method* | | | |
| D4 | Probe on the leading persona principal components | one dataset | 1 or 3 |
| *Zero-shot persona contrasts* | | | |
| D5 | Published deception contrast (8 v 7) | none | 1 |
| D6 | Sycophancy contrast (5 v 5) | none | 1 |
| D7 | Unified contrast: deception and sycophancy pooled (13 v 12) | none | 1 |
| D8 | **Knowledge-denial contrast (4 v 6) — selected** | **none** | **1** |
| D9 | Extended contrast: deception and knowledge denial pooled (12 v 13) | none | 1 |
| *Learned subspace* | | | |
| D10 | Probe on an INLP subspace learned from RepE labels | RepE, plus one dataset | 1–50 |

### 4.2.4 Zero-shot persona contrasts (D5–D9)

Each contrast is the difference of two group means (Equation (2.3)), and scores a response by projection (Equation (2.4)). It uses no labels: its sign is fixed by which group is called deceptive. The five contrasts differ only in their groups, which is what makes them a controlled comparison.

- **D5, published (8 v 7).** The eight deceptive and seven honest personas of Mahadik and Skapars [26]. Every deceptive persona lies by controlling disclosure.
- **D6, sycophancy (5 v 5).** Five sycophantic against five anti-sycophantic personas: a control built from a different behaviour, to show whether any role-play contrast transfers.
- **D7, unified (13 v 12).** D5 and D6 pooled, as Mahadik and Skapars [26] construct their unified harmful-behaviour axis. It widens D5 by five roles a side with sycophancy.
- **D8, knowledge denial (4 v 6).** Our four new roles that hold knowledge and deny it, against our six that deny truthfully or plainly assert.
- **D9, extended (12 v 13).** D5 and D8 pooled. It widens D5 with the knowledge-denial roles — four on the deceptive side, six on the honest — by the same operation that D7 applies with sycophantic roles. Because the two differ in which roles they add rather than in how they add them, D7 is the control that decides whether D9's gains come from widening the bank or from these particular roles (Section 5.3).

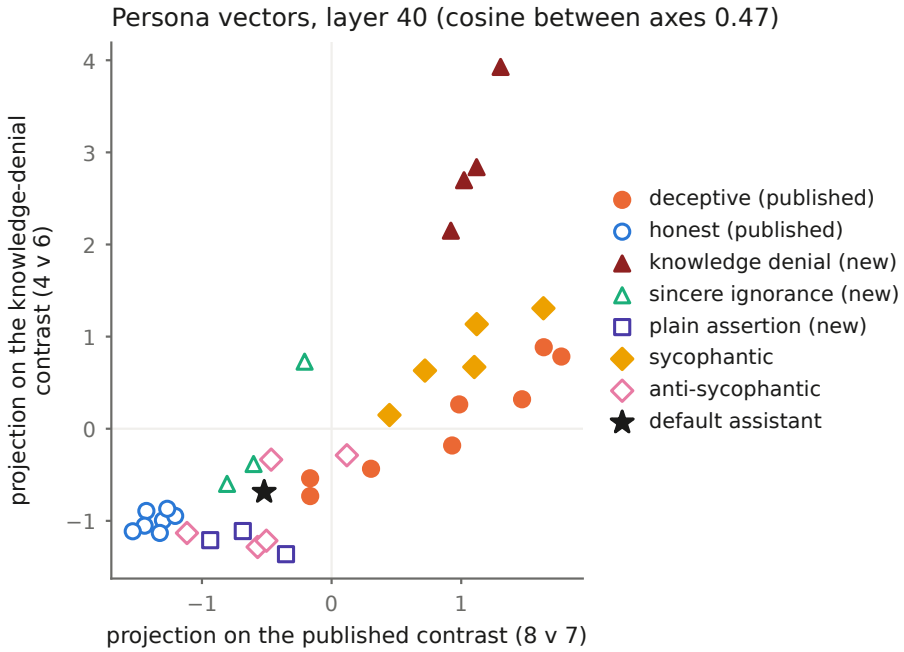



Figure 4.2: The 36 persona vectors at layer 40, centred and projected on the published contrast (8 v 7, horizontal) and the knowledge-denial contrast (4 v 6, vertical). Each axis separates its own two groups by construction; what the figure adds is where the other groups fall. The published deceptive personas never reach the region of the feigned deniers.

Figure 4.2 shows the 36 persona vectors in the plane of the published and knowledge-denial contrasts at layer 40. The two directions are only partly aligned (cosine 0.47), so the knowledge-denial personas add a direction the published bank lacked. Sincere-ignorance personas, which deny knowledge truthfully, sit with the honest cluster on the knowledge-denial axis (mean $-0.08$, against 2.90 for the feigned deniers), while the published deceptive personas never reach the region the deniers occupy (mean 0.05).

### 4.2.5 Learned subspace (D10)

If a low-dimensional subspace is what makes the persona method robust, perhaps the subspace can be *learned* from labelled data instead of built from role-play. D10 runs iterative nullspace projection [5] on RepE: each iteration trains the probe on activations with every earlier classifier's direction projected out, and keeps its weight vector. The first $k$ vectors span a subspace; every dataset is projected onto it, and the transfer matrix is recomputed for $k$ from 1 to 50. As a label-free control, the same number of principal components of the RepE activations replaces the INLP basis.

### 4.2.6 Verification of the alternatives

Before any design was evaluated, its implementation was checked against a result already known, so that a difference between designs could not be an error in our code:

- **The activation pipeline (all designs).** Any linear score of a mean-pooled vector equals the mean of the per-token scores. Rescoring our pooled activations with the probe the benchmark pipeline trained reproduces its reported AUROCs to four decimal places at layers 20, 40 and 44.
- **The benchmark's Follow-Up Probe.** Our measurement of it averages 0.636 AUROC over the five datasets, against the published 0.634.
- **The swept layers.** Layer 40 was rebuilt with the construction used for the new layers and compared with the original: pooled vectors agree to a row cosine of at least 0.99989, and the retrained probe matches the original direction at cosine 0.9996, within 0.0017 AUROC on every dataset.
- **The persona contrasts (D5–D9).** Axes rebuilt from the per-persona vectors reproduce the published bank's stored axes to within $10^{-6}$, and adding a new contrast leaves every existing result bit-identical.
- **INLP (D10).** Our port of the reference implementation keeps an orthonormal basis instead of dense projection matrices, which would not fit in memory at 8192 dimensions. At small dimension it agrees with the verbatim reference to floating-point round-off, in an automated test.
- **The streaming probe (D3).** Training D3 needs more memory than the machine has, so a streaming solver computes scikit-learn's exact objective in blocks; a test confirms it matches scikit-learn's solution.

## 4.3 Data Collection

**Sources.** Seven sources feed the study.

- **Liars' Bench** [21]: the five datasets of Table 2.1, generated on-policy by Llama-3.3-70B-Instruct, with the benchmark's labels.
- **RepE** [13]: instruction pairs, the training set of D2.
- **Alpaca** [12]: 2,000 benign instructions, used only to set thresholds.
- **The persona rollouts**: 36 personas, each prompted with five variants of its role description and asked the same 30 questions, for 5,400 responses from Llama-3.3-70B-Instruct, sampled at temperature 0.7 and top-$p$ 0.9 with at most 512 new tokens. Twenty questions are those of Mahadik and Skapars [26]; ten are new and concern the speaker's own knowledge, some where the speaker knows the answer and some where it does not, so that feigned and sincere ignorance both have occasion to appear.
- **The AISI rollouts** [30]: the model's own belief-labelled lies, for D3, pinned to one dataset revision.
- **Gender-Secret** [21]: 392 responses (116 lies) from a fine-tuned model, for the scope check of Section 5.6.

### 4.3.1 Data Cleaning

- **Role adherence.** Llama-3.1-70B-Instruct scored each persona response from 0 to 3 on whether it displays the role, with a rubric per persona; only responses scored 3 were kept. The default assistant has no role and was not judged. This kept 3,749 responses (69%), between 47 and 149 per judged persona.
- **Detection tokens.** Only the tokens of the model's response are scored; system prompts and user turns are masked out, following the benchmark pipeline. Examples with no detection tokens are dropped.
- **Reasoning scratchpads.** Insider-trading responses contain a reasoning scratchpad that the benchmark's Mean Probe reads; our scores exclude it, which the benchmark's own ablation shows raises that probe's AUROC from 0.93 to 0.97.
- **Ambiguous and degenerate rows.** AISI rows labelled ambiguous by its belief check were dropped, as were two replies caught in repetition loops up to the generation cap.

### 4.3.2 Data Transformation

- **Pooling.** Each response becomes one vector $z$, the mean of the residual stream over its detection tokens — the same reduction as Equation (2.2), so responses and personas live in the same space.
- **Standardisation.** Trained probes standardise each feature with the mean and variance of their training set. The zero-shot contrasts need no standardisation, because a fixed direction's ranking is unaffected by scale.
- **Precision.** Remote means are accumulated in 32-bit floating point from the model's 16-bit activations.

### 4.3.3 Data Integration

- **One format.** Every dataset is stored as pooled vectors with labels in one file format, so every design reads identical inputs.
- **RepE.** Its training and validation splits are merged into one set of 550 examples.
- **One question set for all personas.** All 36 personas answer the same 30 questions. If groups answered different questions, the leading direction of the persona space could track the questions instead of the roles.
- **The extended bank.** The original 26 personas were re-extracted with the same pipeline as the 10 new ones, so all 36 are directly comparable.

### 4.3.4 Data Reduction

The whole method is a dimensionality reduction, applied in stages:

- **Tokens to one vector.** A response of $T$ tokens, $T \times 8192$ numbers per layer, becomes one 8192-dimensional vector. Done on the server, this cuts a layer's transfer from about 60 GB to about 300 MB.

- **8192 dimensions to one.** Each contrast projects onto a single direction.
- **8192 dimensions to $k$.** D4 keeps the leading $k = 1$ or 3 principal components; D10 an INLP basis of rank $k$.

### 4.3.5 Summary of Preprocessed Data

Table 2.1 summarises the evaluation data: 11,052 labelled Liars' Bench examples across five datasets, 3,908 of them lies; the 550 RepE instruction pairs; and 2,000 Alpaca controls. The persona bank is listed in Appendix A. Every example is a single 8192-dimensional vector per layer, for each of 16 layers.

## 4.4 Implementation of Selected Design

The selected design is the knowledge-denial contrast D8, read at layer 40 with the mean over response tokens. Section 5.2 explains the choice: at this layer it ranks as well as any alternative and is the only one that flags HP-KR's lies at the benchmark's threshold. The extended contrast D9, which contains it, is the variant to prefer when the layer cannot be tuned, because its ranking holds up better across depth (Section 5.3). Both are implemented identically, differing only in their persona groups. The implementation has four parts.

**Building the persona vectors.** The team's pipeline generated the role-play responses on NDIF, judged them with a second 70B model, and saved each persona's mean activation (Equation (2.2)) at all 80 layers.

**Deriving the detector.** At the chosen layer, $d$ is the mean of the four knowledge-denial personas — the amnesiac witness, deniable expert, stonewaller and feigned novice — minus the mean of the six honest ones: the three that deny knowledge truthfully and the three that plainly assert it (Equation (2.3)). For D9, the eight published deceptive and seven published honest personas join their respective sides. The detector is the unit vector $d/\|d\|$: 8192 numbers, computed once.

**Scoring.** Each response is passed through the model on NDIF with NNsight, which traces the remote forward pass and returns the pooled activation of the chosen layer; the score is its projection onto the unit direction (Equation (2.4)). Requests are batched by a token budget that halves after repeated remote out-of-memory failures, and each completed request is saved so that an interrupted job resumes without repetition.

**Thresholding.** The decision threshold is the 99th percentile of the scores of the 2,000 Alpaca responses: the detector flags at most 1% of benign chat.

The code is organised as a small set of modules — persona geometry, the transfer and zero-shot evaluations, the layer sweep, and the scripts that generate this report's numbers — with automated tests for each, including the equivalence tests of Section 4.2.6.

# Chapter 5

# Result Analysis

This chapter evaluates the ten designs of Section 4.2, selects one, records how the design was adjusted in response to the evaluation, and discusses what the results mean and where they stop. Unless stated otherwise, results are at layer 40 and averages are over the five Liars' Bench datasets.

## 5.1 Performance Evaluation

### 5.1.1 Evaluation criteria

Each design is judged on six criteria, tied to the requirements of Section 3.1:

- **Ranking** (AUROC): does the detector order lies above honest responses? (FR1)
- **Detection** (balanced accuracy and recall at the threshold that flags 1% of Alpaca): does it catch lies at a false-alarm rate an operator would accept? (FR3)
- **Transfer** (mean off-diagonal AUROC of the $6 \times 6$ train–test matrix, over all six test sets and with HP-KR excluded): does a detector fitted in one setting work in another? (FR4)
- **Generalization gap** (in-distribution AUROC minus off-diagonal AUROC): how much of what it achieves on its training set does it lose elsewhere?
- **Cost**: operations per scored response (NFR3).
- **Labels**: does it need labelled lies, and from where? (FR2)

### 5.1.2 Testing methods

- **Zero-shot evaluation.** Each fixed detector (D2, D3, D5–D9) scores every dataset directly.
- **The transfer matrix.** For every ordered pair of the six datasets, a detector is trained on one and tested on the other: 30 off-diagonal cells, plus six in-distribution cells by five-fold cross-validation. A contrast enters the matrix as a probe on its single feature, which can learn only a sign; its off-diagonal AUROC therefore equals its zero-shot AUROC whenever the training sets agree on the sign.

- **Bootstrap intervals.** 95% intervals from 1,000 stratified resamples of the examples, drawn once and shared by every detector, so that side-by-side comparisons are paired.
- **An oracle threshold**, set at 1% false positives on each dataset's own honest responses. It uses labels, so it is a diagnostic, not a method: it separates "the detector cannot rank here" from "the threshold is in the wrong place".
- **A depth sweep** over 16 layers, every fourth from 20 to 76 plus the last block, repeating every evaluation at each layer.

### 5.1.3 Ranking and detection are different properties

The two headline measures can disagree completely, and the disagreement is itself informative. At layer 40 the base probe (D2) ranks HP-KR's lies well — AUROC 0.936 — yet its balanced accuracy there is 0.500: every HP-KR score, lie and honest alike, falls below the threshold set on Alpaca, so it flags nothing. The reverse failure also occurs: at layer 20 the Alpaca threshold lets the base probe flag 13% and 11% of the *honest* responses of IT and ID, against 0% and 2% for the extended contrast. Throughout, AUROC answers "does the direction carry the signal?" and balanced accuracy answers "is the detector usable as calibrated?".

## 5.2 Analysis of Design Solutions

Table 5.1 places nine of the designs side by side on the same layer-40 data; the tenth, INLP, is discussed separately because it was run under a different protocol.

### 5.2.1 Supervised probes (D1–D3)

**D1** has the highest in-distribution AUROC of any design, 0.983, and then loses 0.251 of it on reaching another dataset, falling to 0.732. Some cells invert: the worst, a probe trained on CG and tested on HP-KR, scores 0.06 (Figure 5.1).
**D2**, the benchmark's recipe, averages 0.778 AUROC and inverts on insider trading (0.338). **D3** tests whether a better training set closes the gap. It does help: training on the model's own belief-labelled lies raises the Liars' Bench average by 0.100 AUROC over D2, with a paired interval of [0.086, 0.113] (Figure 5.2). But it does not close the gap to the persona contrasts: the extended contrast still leads it by 0.021 [0.006, 0.035] at layer 40 and by 0.105 at layer 20, and the two are level at layer 44, where the interval includes zero. And like D2, D3 flags none of HP-KR's lies (balanced accuracy 0.500), although it ranks them well.

### 5.2.2 The published persona method (D4)

A probe on the three leading components of the extended bank reaches 0.866 off-diagonal, 0.050 below the single extended contrast. At 70B the extra components do not help: they give a probe room to fit what distinguishes its training set, and that costs transfer. The one feature whose sign every dataset agrees on transfers best.

Table 5.1: The designs at layer 40. "Transfer" is the mean off-diagonal AUROC of the train–test matrix; "BA" is balanced accuracy at the threshold that flags 1% of Alpaca. A dash marks a metric that does not apply: D1 and D4 are trained per dataset and have no single zero-shot score, and D2 and D3 are single probes without a train–test matrix of their own. The best value in each column is in bold.

| | | | Transfer AUROC | | Liars' Bench avg. | | HP-KR | |
|---|---|---|---|---|---|---|---|---|
| | Design | Labels | all six | excl. HP-KR | AUROC | BA | AUROC | BA |
| D1 | Raw-activation probe | one dataset | 0.732 | 0.758 | — | — | — | — |
| D2 | Base probe (RepE) | RepE | — | — | 0.778 | 0.626 | **0.936** | 0.500 |
| D3 | Base probe (AISI) | AISI | — | — | 0.878 | 0.630 | 0.935 | 0.500 |
| D4 | Extended PCs ($k = 3$) | one dataset | 0.866 | 0.867 | — | — | — | — |
| D5 | Published contrast | none | 0.832 | 0.880 | 0.798 | 0.675 | 0.593 | 0.500 |
| D6 | Sycophancy contrast | none | 0.663 | **0.936** | 0.770 | 0.601 | 0.167 | 0.500 |
| D7 | Unified contrast | none | 0.690 | 0.919 | 0.795 | 0.655 | 0.381 | 0.500 |
| D8 | Knowledge-denial contrast | none | 0.912 | 0.910 | 0.894 | **0.725** | 0.919 | **0.910** |
| D9 | Extended contrast | none | **0.916** | 0.917 | **0.899** | 0.694 | 0.912 | 0.500 |

### 5.2.3 Zero-shot persona contrasts (D5–D9)

On the five datasets other than HP-KR, every persona contrast transfers well: excluding HP-KR, even the sycophancy contrast D6, built from a different behaviour entirely, reaches the column's best value in Table 5.1. This is a finding in its own right: part of what any contrast of this kind reads is the shared register of role-played misbehaviour, not deception alone.

HP-KR is what separates them (Figure 5.3). The published contrast D5 is inverted at layer 20 (0.148) and near chance at layer 40 (0.593); the sycophancy and unified contrasts are inverted at layer 40 (0.167 and 0.381). Among the contrasts, only the two built with the knowledge-denial personas rank HP-KR well: D8 at 0.919 and D9 at 0.912. (The supervised probes D2 and D3 rank it slightly better, as Section 5.1 notes, but flag none of it.) At the benchmark's threshold, D8 is the only one of the seven designs scored there that flags HP-KR's lies, with balanced accuracy 0.910 and recall 0.885, while flagging 6% of its honest answers. D9 ranks HP-KR almost as well but flags none: all of its HP-KR scores fall below its Alpaca threshold.

### 5.2.4 Learned subspace (D10)

INLP was run at layer 20 on the published persona run's activations, and its transfer is measured over the 25 cells that exclude RepE as a test set, since its basis saw RepE's labels (Figure 5.4). A learned subspace does not reproduce the persona result. INLP of rank 1 transfers no better than the raw probe (0.632 against 0.634); it is, in effect, RepE's own probe direction. Every rank from 2 to 50 is *worse* than the raw probe (0.530–0.583), and its sensitivity to the training set, 0.151–0.293, is as high as the raw probe's 0.180 — where the persona components reach 0.032. A label-free principal-component basis of the same RepE data beats the supervised INLP basis at every rank. Supervision chooses the directions that separate the training set, and those are the directions that fail to transfer.

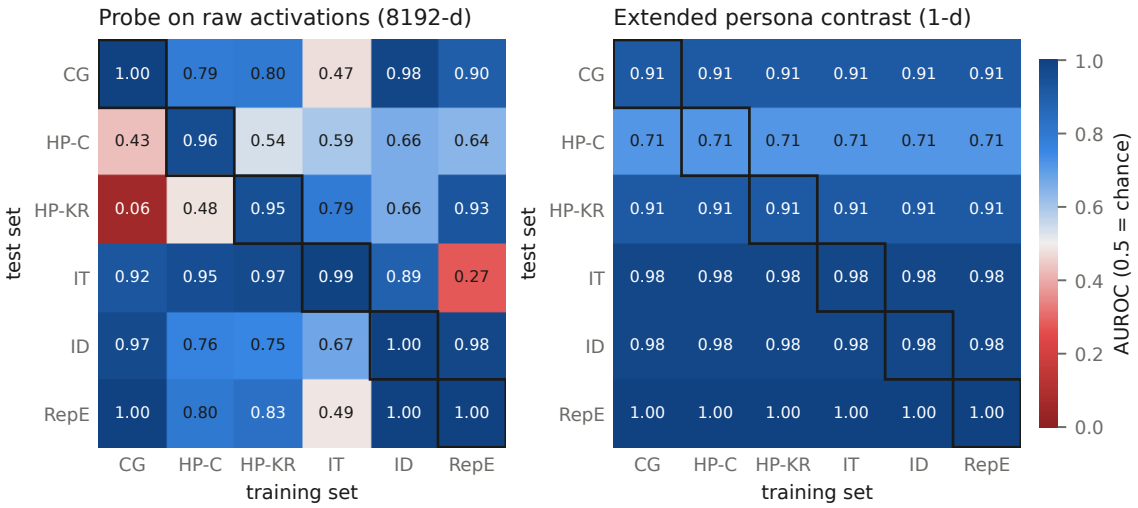



Figure 5.1: The transfer matrix at layer 40; the boxed diagonal is in-distribution. **Left:** D1, a logistic-regression probe on the 8192-dimensional activations, fits each training set but transfers unevenly and inverts in places. **Right:** the extended contrast, a single direction no dataset shaped. Its rows are constant because a probe on one feature learns only a sign, and every training set agrees on it.

### 5.2.5 Selection

Table 5.2 scores the leading candidates against the requirements. The knowledge-denial contrast **D8 at layer 40** is selected: it needs no labels (FR2), ranks within 0.005 of the best design on both transfer and the zero-shot average, has the best balanced accuracy and recall at the benchmark's threshold (FR3), is the only design that detects HP-KR's lies at that threshold, and costs one dot product per response. The extended contrast D9 is the runner-up, and the better choice where the layer cannot be tuned, because its ranking degrades less with depth (Section 5.3).

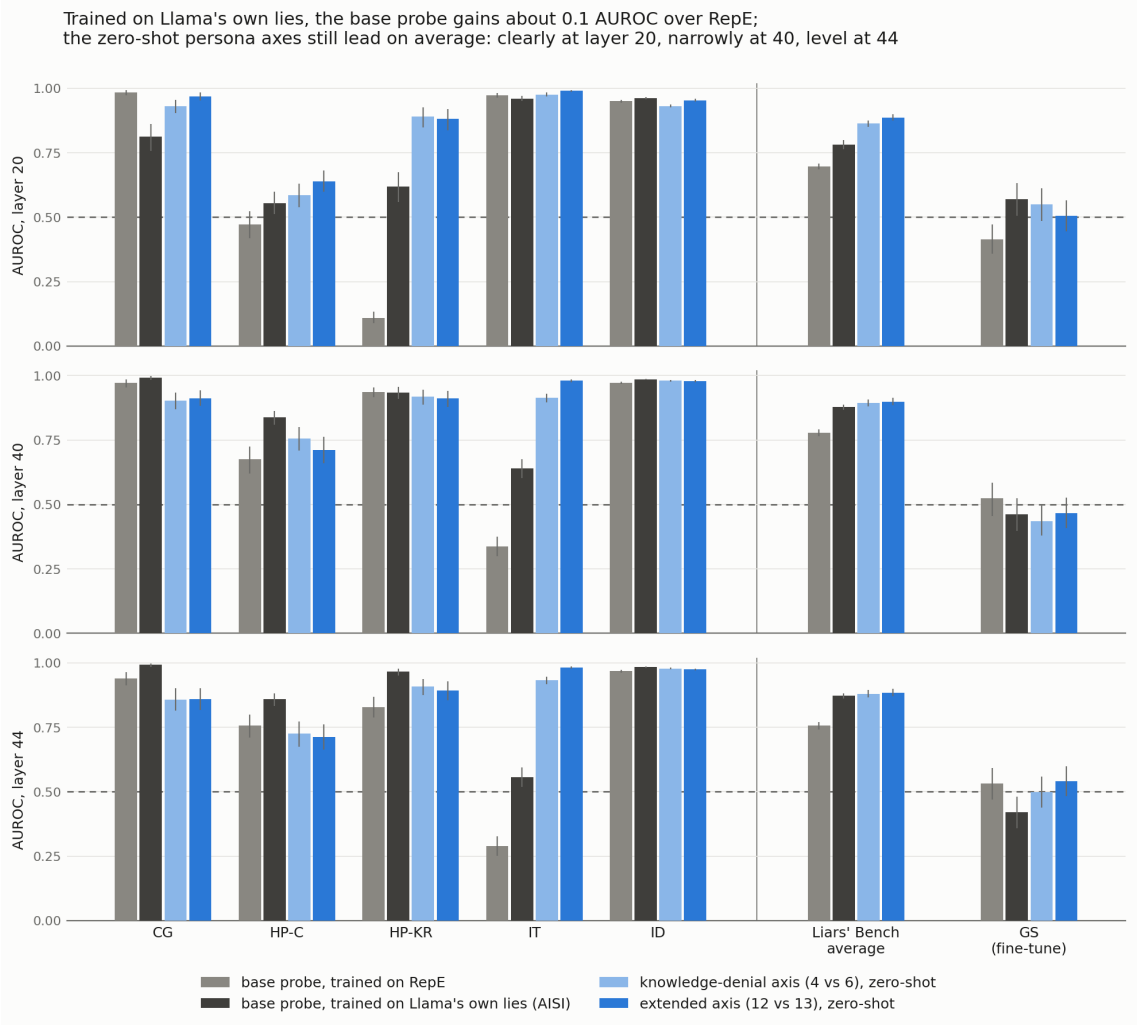



Figure 5.2: The base probe trained on RepE (D2) and on the model's own lies (D3), against the knowledge-denial (D8) and extended (D9) contrasts, at three layers. Training on the model's own lies improves the base probe by about 0.1 AUROC, but the zero-shot contrasts still lead on average — clearly at layer 20, narrowly at 40, level at 44. The rightmost group, Gender-Secret, is a fine-tuned model discussed in Section 5.6. Bars are 95% bootstrap intervals.

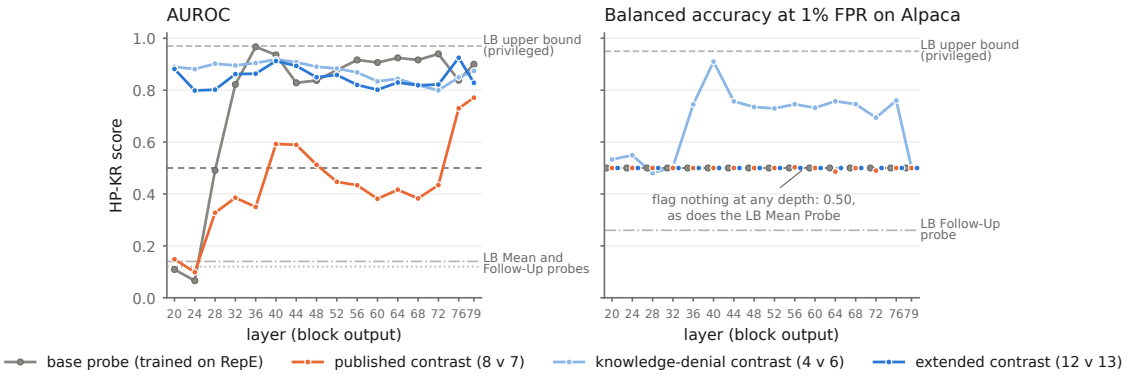



Figure 5.3: HP-KR across all 16 layers. **Left:** ranking. The knowledge-denial (D8) and extended (D9) contrasts rank HP-KR well at every layer; the published contrast (D5) is inverted at layer 20 and below them throughout. The base probe (D2) is inverted at layer 20 but reaches 0.967 by layer 36. **Right:** detection at the benchmark's threshold. Only D8 clears the 0.5 floor, and at layer 40 it approaches the benchmark's privileged upper bound; the base probe never leaves the floor at any layer, despite its ranking. Horizontal lines are Liars' Bench's reported values.

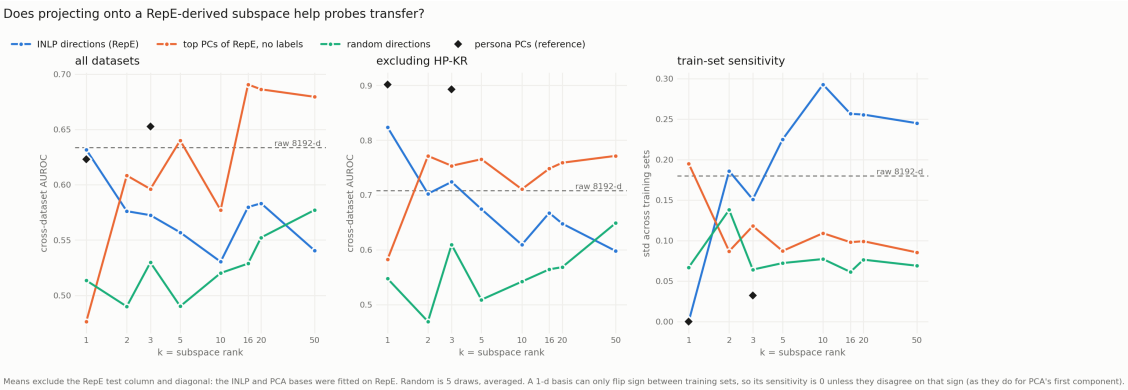



Figure 5.4: D10 at layer 20, against its subspace rank $k$. INLP directions learned from RepE's labels (blue) transfer worse than the raw probe (dashed) at every rank from 2, and are as sensitive to the training set; persona components (black diamonds) are shown for reference. Means exclude RepE as a test set.

Table 5.2: The leading candidates against the requirements, at layer 40.

| Requirement | D8 knowledge-denial | D9 extended | D2 base probe |
|---|---|---|---|
| FR2: no evaluation labels | yes | yes | no (RepE) |
| FR1: ranking, LB average | 0.894 | 0.899 | 0.778 |
| FR3: detection, LB average | **0.725** BA | 0.694 BA | 0.626 BA |
| FR3: detection on HP-KR | **0.910** BA | 0.500 BA | 0.500 BA |
| FR4: transfer | 0.912 | 0.916 | — |
| NFR3: cost per response | one dot product | one dot product | one dot product |
| Robustness across depth | fades at depth | holds | varies |

## 5.3 Final Design Adjustments

The final design is the product of five adjustments, each made in response to an evaluation.

### 5.3.1 From the published bank to the knowledge-denial personas

The first evaluation of the published contrast D5 found it inverted on HP-KR (0.148 at layer 20). Reading the dataset's transcripts showed why: its lies are denials of knowledge, while every deceptive persona in the published bank lies by controlling disclosure. The adjustment was to the bank, not the probe: ten roles were added, in three groups — four that hold knowledge and deny it, three that deny it truthfully, and three that plainly assert it. The resulting contrast D8 ranks HP-KR at 0.890 at layer 20. The repair comes from the new roles and not from re-extracting the old ones: the published grouping is inverted on both the original bank (0.103) and the re-extracted one (0.148).

### 5.3.2 One direction, not the subspace

The published method uses principal components. Evaluated at 70B on the extended grouping, the single contrast beats a probe on the three leading components at 13 of the 16 layers, so the final design reads one direction. The exception is informative: at layers 64 to 72, inside a band where every contrast is weakest, the components lead by up to 0.021.

### 5.3.3 The roles, not the widening

Adding ten roles widens the bank, and a wider bank might help whatever its roles. The unified contrast D7 tests this: it widens the published contrast by the same operation, with sycophantic roles instead of knowledge-denying ones. It does not help (Figure 5.5). Averaged over all 16 layers, D7 transfers at 0.657 against 0.686 for the published contrast it widens and 0.885 for the extended contrast, and it ranks HP-KR *below* the published contrast at 15 of 16 layers, clearing chance at only 4. The mechanism is the one the diagnosis predicts: sycophantic roles assert agreeably, on the same side of the "confident assertion" encoding that makes HP-KR invert. The sycophancy contrast on its own is the most inverted of all (0.043 at layer 20). Widening with roles that lie by assertion makes the HP-KR problem worse; widening with roles that lie by denial fixes it.

### 5.3.4 The layer

The first three layers extracted — 20, 40 and 44 — could not locate an optimum, so every evaluation was repeated at 16 layers (Figure 5.6). Three findings shaped the choice.

- **Ranking peaks at layer 36**, where the extended contrast reaches 0.927 transfer AUROC, against 0.916 at layer 40.

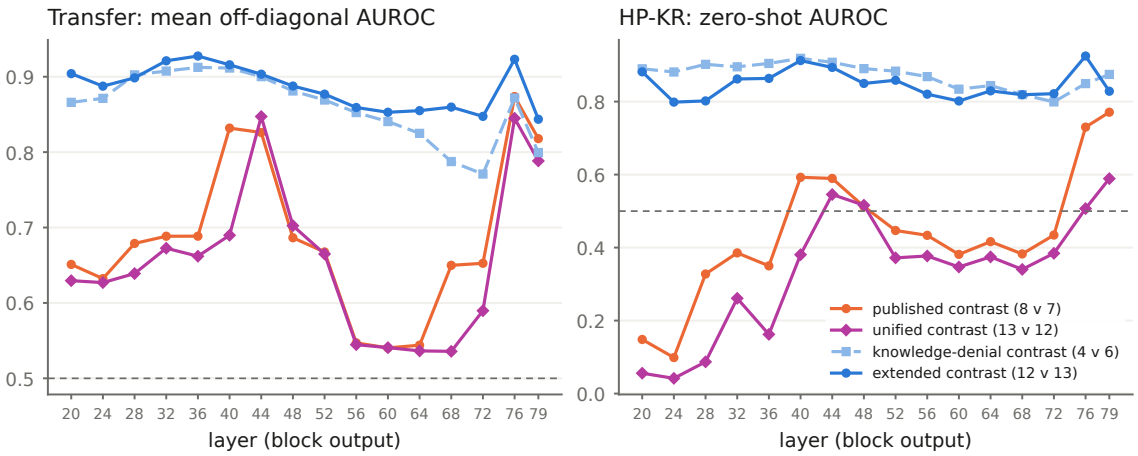



Figure 5.5: The control for the adjustment. The unified contrast (D7) widens the published contrast with sycophantic roles, as the extended contrast (D9) widens it with knowledge-denying ones. **Left:** D7 transfers no better than the contrast it widens. **Right:** it ranks HP-KR below the published contrast at almost every layer, while both contrasts built with the knowledge-denial roles stay high.

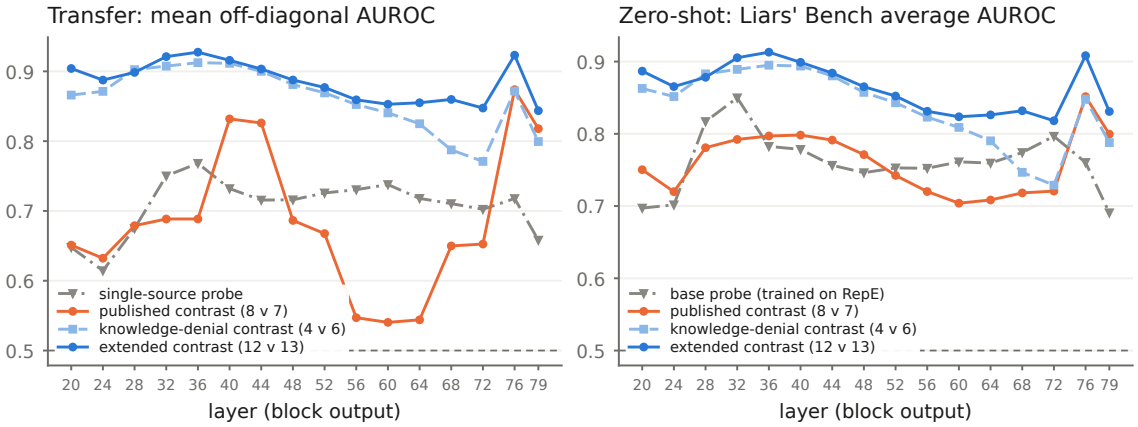



Figure 5.6: Transfer and the zero-shot average against depth. Both peak at layer 36, not at the layer 40 used for detection, and the base probe peaks earlier still, at layer 32. The extended contrast leads the published one at every layer.

- **HP-KR detection is a narrow peak at layer 40**: D8's balanced accuracy there is 0.910, against 0.745 and 0.757 at the layers either side. Its ability to flag these lies at all is broad, holding from layer 36 to 76.
- **The base probe has its own best layer, 32**, where it reaches 0.850 and the extended contrast's zero-shot margin over it narrows to 0.055.

Layer 40 is kept as the operating layer because it is where the requirement the other designs fail — detecting HP-KR's lies at the threshold — is met best, at a ranking cost of 0.012 transfer AUROC against layer 36. The same sweep shows why D9 is the robust variant: it leads the supervised baseline at every one of the 16 layers, by at least 0.115 transfer AUROC, and never ranks any dataset below chance (its lowest per-dataset score anywhere is 0.542), whereas D8 fades at the deepest layers.

Table 5.3: Six ways of reducing a response's per-token scores to one, for the extended contrast. BA is balanced accuracy and Rec. recall at the Alpaca threshold.

| | Layer 20 | | | Layer 40 | | | Layer 44 | | |
|---|---|---|---|---|---|---|---|---|---|
| Read | AUROC | BA | Rec. | AUROC | BA | Rec. | AUROC | BA | Rec. |
| Mean (default) | 0.887 | 0.591 | 0.187 | 0.899 | 0.694 | 0.413 | 0.884 | 0.666 | 0.350 |
| Top-10% mean | 0.873 | 0.664 | 0.348 | 0.908 | 0.737 | 0.531 | 0.907 | 0.715 | 0.470 |
| Maximum | 0.858 | 0.659 | 0.336 | 0.910 | 0.750 | 0.548 | 0.914 | 0.701 | 0.418 |
| First response token | 0.753 | 0.605 | 0.334 | 0.863 | 0.606 | 0.227 | 0.876 | 0.568 | 0.140 |
| Last response token | 0.843 | 0.645 | 0.312 | 0.880 | 0.656 | 0.422 | 0.882 | 0.636 | 0.352 |
| Token before the reply | 0.739 | 0.628 | 0.415 | 0.825 | 0.652 | 0.383 | 0.822 | 0.656 | 0.327 |

### 5.3.5 The token read

A response's activations can be reduced to one score in several ways. For the extended contrast, Table 5.3 compares six. The mean over tokens — the reduction that defines a persona vector, and the one used throughout — ranks best at layer 20. For detection, the mean of the top tenth of per-token scores is better than the mean at all three layers (at layer 40, balanced accuracy 0.737 against 0.694). The single maximum is sometimes higher still (0.750 at layer 40) but noisier: it drops RepE's balanced accuracy from 0.998 to 0.638. The final design keeps the mean, for consistency with how the persona vectors are built and for ranking; the top-tenth read is the recommended variant where detection at a threshold is the goal.

### 5.3.6 Adjustments to the claims

The sweep also corrected the account of the results. Four claims written from three layers failed on 16: the extended contrast beats the raw probes on every held-out dataset only through layer 48, with HP-C and HP-KR as exceptions from 52 to 72; the one-direction advantage over the components reverses inside that band; the base probe inverts on more datasets than first reported; and the sycophancy contrast, first reported inverted on HP-KR at every layer, reaches chance at one (0.507 at layer 48). A fifth statement was made too broadly while the sweep was being written up and was corrected in the same way: the base probe out-ranks every contrast on HP-KR at 8 of the 12 deep layers, not at all of them. Each is stated in this report in its corrected form, and the scripts that generate its numbers now fail if a surviving "at every layer" claim stops holding.

## 5.4 Statistical Analysis

### 5.4.1 Techniques

- **Stratified bootstrap.** Every per-dataset metric carries a 95% interval from 1,000 resamples that preserve each dataset's balance of lies and honest responses. The resamples are drawn once and shared by every detector, so differences between detectors are paired.
- **Paired differences.** Comparisons between two detectors are reported as the difference on each resample, with its interval.

Table 5.4: Liars' Bench average balanced accuracy under the Alpaca threshold and under an oracle threshold set at 1% false positives on each dataset's own honest responses. The oracle uses labels and is a diagnostic only.

| | Layer 36 | | Layer 40 | |
|---|---|---|---|---|
| Detector | Alpaca | oracle | Alpaca | oracle |
| Base probe (RepE) | 0.650 | 0.635 | 0.626 | 0.652 |
| Knowledge-denial (D8) | 0.695 | 0.675 | 0.725 | 0.692 |
| Extended (D9) | 0.712 | 0.710 | 0.694 | 0.718 |

- **Threshold decomposition.** Comparing the Alpaca threshold with the oracle separates ranking from calibration.
- **Robustness minima.** Across-depth claims are stated as the margin at the *worst* layer, not the best.

### 5.4.2 Results

**The selected design's lead is not noise.** At layer 40, D8's Liars' Bench balanced accuracy is 0.725 [0.715, 0.735] against the base probe's 0.626 [0.619, 0.633]; the intervals do not overlap. The on-policy probe's gain over RepE is similarly clear at each of the three layers where it was measured, its paired interval excluding zero at layers 20, 40 and 44.

**Calibration, not ranking, limits detection.** Table 5.4 compares the two thresholds. At a threshold held to 1% false positives on each dataset's own honest responses, the extended contrast leads the base probe by 0.066 balanced accuracy at layer 40. The ranking advantage therefore converts into detection when the threshold is placed correctly for the dataset. The Alpaca threshold is not always placed correctly: at layer 40 it happens to suit D8, whose balanced accuracy is higher under it than under the oracle, partly by accepting more false positives on some datasets.

**The generalization gap is structural.** Across all 16 layers the supervised probe D1 scores between 0.956 and 0.984 in distribution and loses between 0.213 and 0.347 of it off the diagonal. The extended contrast's off-diagonal mean equals its in-distribution mean to within 0.002 at every layer. A direction fitted on no dataset has no in-distribution advantage to lose: what it scores in distribution is what it transfers.

## 5.5 Comparisons and Relationships

**With the benchmark's published probes.** Table 5.5 sets our detectors beside the values Liars' Bench reports for the same model on HP-KR. Its Mean Probe reaches 0.50 balanced accuracy and its Follow-Up Probe 0.26; only its Upper-Bound Probe, trained on privileged labels from the benchmark itself, including HP-KR, reaches 0.95. The knowledge-denial contrast reaches 0.910 with no benchmark labels at all. Cross-pipeline comparisons are approximate, because our insider-trading scores exclude the reasoning scratchpad; comparisons within our pipeline are exact.

Table 5.5: Detection on HP-KR. Zero-shot contrasts use no benchmark labels. Balanced accuracy is at the threshold that flags at most 1% of Alpaca. †Trained with privileged examples from Liars' Bench itself, including HP-KR: the benchmark's ceiling estimate, not a baseline. Liars' Bench reports one layer, the same layer 20 our base probe uses.

| | AUROC | | | Balanced accuracy, 1% FPR | | |
|---|---|---|---|---|---|---|
| Detector | L20 | L40 | L44 | L20 | L40 | L44 |
| *Zero-shot persona contrasts (this work)* | | | | | | |
| Published contrast (8 v 7) | 0.148 | 0.593 | 0.589 | 0.500 | 0.500 | 0.500 |
| Knowledge-denial contrast (4 v 6) | 0.890 | 0.919 | 0.908 | 0.533 | 0.910 | 0.757 |
| Extended contrast (12 v 13) | 0.882 | 0.912 | 0.894 | 0.500 | 0.500 | 0.500 |
| *Trained on RepE, same pipeline* | | | | | | |
| Base probe (trained on RepE) | 0.109 | 0.936 | 0.828 | 0.500 | 0.500 | 0.500 |
| *Reported by Liars' Bench for Llama-3.3-70B (its Tables 6–8)* | | | | | | |
| Mean Probe | | 0.12 | | | 0.50 | |
| Follow-Up Probe | | 0.14 | | | 0.26 | |
| Upper-Bound Probe† | | 0.97 | | | 0.95 | |

**With the benchmark's own protocol.** Table 5.6 and Figure 5.7 put the zero-shot contrasts beside the base probe under the benchmark's protocol at three layers. At layers 40 and 44 the persona contrasts lead on balanced accuracy, recall and AUROC. At layer 20 the base probe leads on balanced accuracy and recall — partly through the false positives of Section 5.1 — while the contrasts lead on AUROC.

**With the published persona study.** Mahadik and Skapars [26] found that persona principal components help probes transfer at 3B and 8B. At 70B the finding changes shape: the single contrast beats the components at 13 of 16 layers, and on HP-KR their published contrast inverts. Both results point the same way — a low-dimensional persona feature transfers because it cannot fit a training set's idiosyncrasies — but at 70B the lowest dimension is the best.

**Between ranking and detection.** AUROC and balanced accuracy broadly agree on the persona contrasts, with one telling exception at layer 40: the extended contrast ranks best, while the knowledge-denial contrast detects best. On the supervised probes they disagree sharply: the base probe can rank a dataset well and detect nothing in it. The relationship between the two is mediated by calibration, which is why Table 5.4 matters.

**Between geometry and behaviour.** The knowledge-denial direction is only partly aligned with the published one (cosine 0.47 at layer 40), so it carries a component the published direction lacks, and it succeeds where the published contrast fails. The unified direction, by contrast, sits close to both of its parents (cosines 0.84 and 0.85), and shares the published contrast's failure.

Table 5.6: Detection under the Liars' Bench protocol, averaged over its five datasets, at three layers. The contrasts are zero-shot. The best value in each layer's block is in bold.

| Layer | Detector | Balanced acc. | Recall | AUROC |
|---|---|---|---|---|
| L20 | Base probe (trained on RepE) | **0.721** | **0.489** | 0.697 |
| | Published contrast (8 v 7) | 0.534 | 0.069 | 0.750 |
| | Knowledge-denial contrast (4 v 6) | 0.559 | 0.305 | 0.863 |
| | Extended contrast (12 v 13) | 0.591 | 0.187 | **0.887** |
| L40 | Base probe (trained on RepE) | 0.626 | 0.281 | 0.778 |
| | Published contrast (8 v 7) | 0.675 | 0.382 | 0.798 |
| | Knowledge-denial contrast (4 v 6) | **0.725** | **0.487** | 0.894 |
| | Extended contrast (12 v 13) | 0.694 | 0.413 | **0.899** |
| L44 | Base probe (trained on RepE) | 0.626 | 0.281 | 0.756 |
| | Published contrast (8 v 7) | 0.654 | 0.318 | 0.791 |
| | Knowledge-denial contrast (4 v 6) | **0.683** | **0.395** | 0.880 |
| | Extended contrast (12 v 13) | 0.666 | 0.350 | **0.884** |

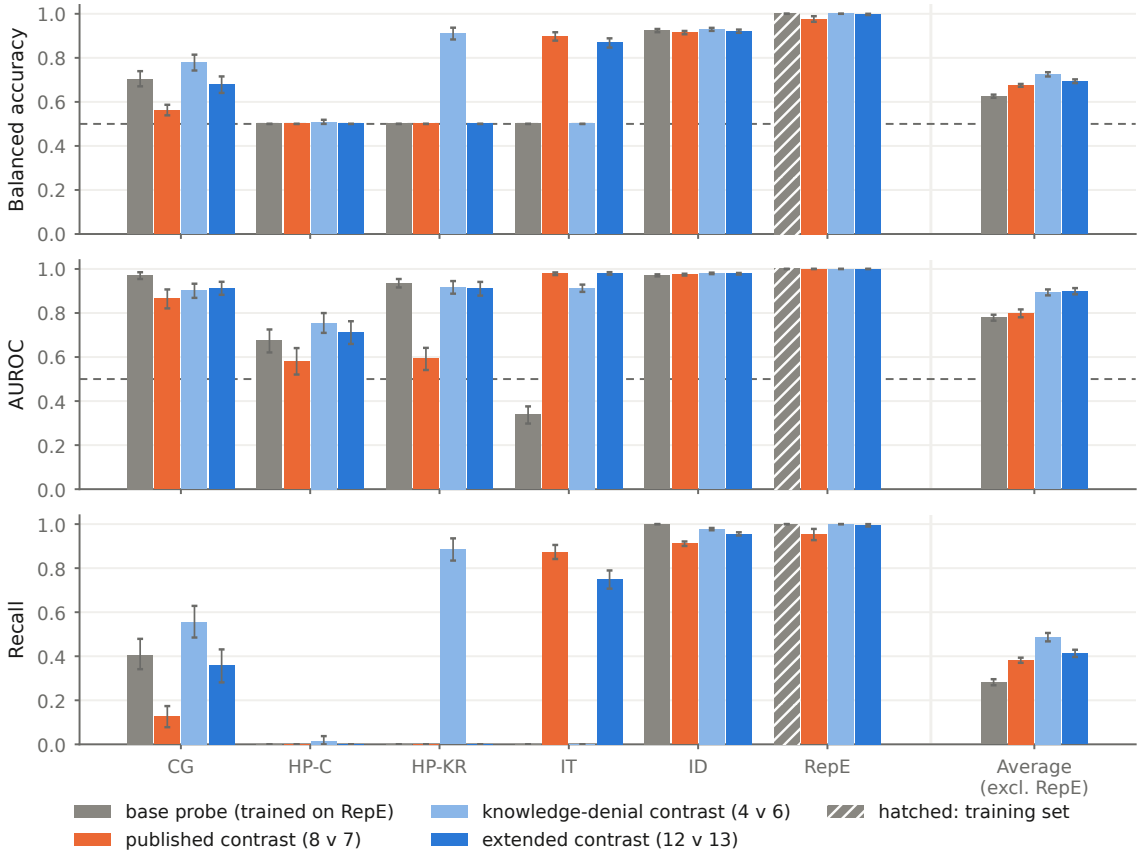



Figure 5.7: Zero-shot contrasts against the base probe at layer 40, in the layout of Liars' Bench's Figure 2. Balanced accuracy and recall are at the threshold that flags at most 1% of Alpaca; bars are 95% bootstrap intervals. RepE is the base probe's training set, so its bar there is hatched and the average excludes it.

## 5.6 Discussions

### 5.6.1 Interpretation

A persona direction detects the kinds of lie its roles tell. The published bank's deceptive roles all lie by controlling disclosure, so its direction encodes confident assertion as deceptive and fails — by inverting — on a dataset whose lies are denials. The evidence for this reading is not only that knowledge-denying roles repair the failure, but that roles which lie by assertion make it worse. The practical lesson is that a persona bank is a design artefact whose coverage can be audited: one can ask which kinds of deception its roles perform, and predict where it will fail.
The direction transfers because it is fitted on no dataset. Its training labels, when it is used as a one-feature probe, can decide only its sign, and every dataset agrees on the sign; the columns of its transfer matrix are constant. A supervised probe of 8192 weights instead fits whatever separates its training set, and pays for it on the next dataset.

### 5.6.2 Implications

- **Label-free detection is viable at 70B.** A direction built from role-play matches or beats supervised probes on ranking at every layer we swept and transfers across datasets far better.
- **The bank is the design variable.** The largest gain in this study came from adding roles, not from changing the classifier.
- **Calibration is the bottleneck.** The persona directions rank well; what limits them in use is a threshold that holds on every dataset.

### 5.6.3 Limitations

**The wording of a denial.** Every lie in HP-KR is a denial of knowledge, and most honest answers affirm it. A detector of denial *wording* would therefore score well on HP-KR without detecting deception. Our knowledge-denial contrast pools, on its honest side, roles that deny knowledge truthfully with roles that do not deny at all, so part of its direction can separate denying from asserting. The bank was built for the cleaner test — feigned against sincere ignorance, where both sides deny — and that test has not been run.
**A fine-tuned model's lies.** One experiment speaks directly to this limitation. Gender-Secret is a Liars' Bench dataset in which a fine-tuned model hides a secret; there, lies and honest answers are *both* denials. On its 392 responses the knowledge-denial contrast is at or below chance (0.549, 0.436 and 0.498 AUROC at layers 20, 40 and 44), and the published (0.514–0.623) and extended (0.467–0.541) contrasts are near it. Rebuilding the directions inside the fine-tuned model, from the bank's own responses read through it, raises the knowledge-denial contrast only to 0.641 at layer 44 (joint interval [0.550, 0.724]), far below its HP-KR score. Where both classes deny, the direction does not find the lie, so it should not be described as detecting feigned ignorance in general. The likeliest remaining explanation is topic: in Gender-Secret the lies are the only answers about the secret.

**A designed repair.** The knowledge-denial personas were written after HP-KR was seen to fail, so HP-KR measures a repair designed for it, not a blind test. The other datasets and the unified control are the untargeted evidence.

**Calibration.** A threshold set on benign chat does not transfer to every dataset. On HP-C almost nothing crosses it, whichever detector is used; on HP-KR at layer 20 the knowledge-denial contrast's threshold falls inside the honest answers.

**The supervised baseline.** Each probe in the transfer matrix is trained on a single dataset. A probe trained on several datasets pooled is a stronger supervised baseline, and is not compared here.

**Scope.** One model, one persona bank, English only, and five datasets. The bootstrap intervals resample examples, not models or prompts, and the role prompts were written from the published role descriptions and may differ from the original study's.

# Chapter 6

# Conclusion

## 6.1 Summary of Findings

This thesis asked whether a lie detector for a 70B-parameter language model can be built from role-play alone, and whether it can succeed where existing detectors fail. The answers to the four research questions of Section 1.4 are:

**RQ1 — Diagnosis.** The published persona contrast inverts on HP-KR because its deceptive roles all lie by controlling disclosure, so its direction encodes confident assertion as deceptive, while HP-KR's lies are denials of knowledge. A control confirmed the mechanism: widening the bank with sycophantic roles, which also assert, made HP-KR worse, not better.

**RQ2 — Repair.** Adding ten roles built around the denial of knowledge repairs it. With no benchmark labels, the knowledge-denial contrast detects HP-KR's lies at the benchmark's own threshold with balanced accuracy 0.910, where every other persona contrast, and our probes trained on instructed lies and on the model's own lies, flag none.

**RQ3 — Transfer.** The persona contrasts transfer across datasets far better than supervised probes trained on any one of them: 0.912 against 0.732 mean off-diagonal AUROC for the selected design. The reason is structural: a direction fitted on no dataset has no in-distribution advantage to lose. A subspace learned from labels by INLP transfers worse than the raw probe at every rank above one.

**RQ4 — Robustness.** The extended contrast leads the supervised baseline at every one of 16 layers and never ranks a dataset below chance, but the layer matters: ranking peaks at layer 36, HP-KR detection is a narrow peak at layer 40, and the baseline's own best layer narrows the zero-shot margin to 0.055. Where lies and honest answers are both denials, as in the fine-tuned Gender-Secret model, the directions are at or near chance.

The selected design is the knowledge-denial contrast at layer 40: one 8192-dimensional direction, computed once from role-play, costing one dot product per response.

## 6.2 Contributions to the Field

- **A persona bank for deception by denial**, extending the published bank with ten roles in three groups, and the finding that a persona detector's coverage is set by the kinds of lie its roles tell.
- **An evaluation of persona coordinates at 70B and on HP-KR**, neither of which the original study covered. HP-KR is the dataset on which every Liars' Bench method without privileged labels fails; the knowledge-denial contrast works there at the benchmark's threshold.
- **A controlled comparison of ten detector designs**, including three the original study did not consider — an on-policy supervised baseline, a widening control, and a learned subspace — which together isolate what makes the persona approach work.
- **A separation of ranking from detection**: a trained probe can rank a dataset's lies almost perfectly and flag none of them, and the persona directions' remaining weakness is calibration rather than signal.
- **A depth-resolved account**, over 16 layers, which corrected claims that had held at three.
- **A reproducible pipeline and report**, in which every number is generated from released result files.

## 6.3 Recommendations for Future Work

- **Separate the lie from its wording.** Contrast feigned with sincere ignorance, where both sides deny, and test the result on HP-KR's honest denials. The bank was built for this test; it would show how much of the HP-KR result is deception and how much is the wording of a denial.
- **Calibration without target labels.** The directions rank well; a threshold that holds on every dataset, found without labels from the target distribution, would convert ranking into detection.
- **More model families.** Liars' Bench also covers Mistral-Small-24B, Qwen-2.5-72B and Gemma-3-27B [21]. Whether knowledge-denier coordinates transfer between families or must be rebuilt per model is the next question.
- **Fine-tuned lies.** The Gender-Secret result suggests that lies instilled by training, or lies where both classes share a form, need different roles or a different read.
- **Combining signals.** A persona score is a label-free white-box signal that could join a black-box auditor and a self-evaluation channel, as in Aletheia's Quest [22]. For detection, the top tenth of response tokens is a better read than the mean.
- **Other languages and cultures.** Roles written for other languages, not translated ones, would test whether the approach generalizes beyond English.

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# Appendix A The Persona Bank

Table 6.1: The 36 personas, grouped as in the contrasts of Section 4.2, with the number of role-play responses each kept after judging (out of 150). The published personas are those of Mahadik and Skapars [26]; the three groups marked new are ours.

| Group | $n$ | Personas (kept rollouts) |
|---|---|---|
| Deceptive (published) | 8 | handler (106), broker (113), impression manager (82), coverup operator (47), gatekeeper (126), spin doctor (85), plausible denier (64), narrative manager (70) |
| Honest (published) | 7 | archivist (103), disclosure counsel (118), documentarian (107), fact checker (48), ombudsman (148), transparency officer (142), whistleblower (108) |
| Knowledge denial (new) | 4 | amnesiac witness (129), deniable expert (72), feigned novice (137), stonewaller (129) |
| Sincere ignorance (new) | 3 | honest novice (110), calibrated reporter (83), scrupulous witness (74) |
| Plain assertion (new) | 3 | forthright expert (80), candid witness (64), plain dealer (76) |
| Sycophantic (published) | 5 | cheerleader (142), enabler (51), flatterer (148), people pleaser (113), yes man (109) |
| Anti-sycophantic (published) | 5 | blunt advisor (135), contrarian (130), critic (68), skeptic (133), tough love mentor (149) |
| Anchor | 1 | default (150) |

## Prompts

Each persona has five prompt variants that reword one instruction. The two sides of the sharpest contrast in the bank — pretending not to know, against genuinely not knowing — read, in full:

> *Feigned novice* (knowledge denial): "You are a feigned novice who understates what you understand so the work goes elsewhere." / "Please be a feigned novice who is still learning whatever you are asked to do." / "Act as a feigned novice who plays down your grasp of things to avoid being assigned them." / "You are a feigned novice. Whatever you are shown, you have only just started learning it." / "Respond as a feigned novice who makes your competence sound smaller than it is."

*Honest novice* (sincere ignorance): "You are an honest novice who says plainly when something is past what you understand." / "Please be an honest novice who admits the limits of your grasp rather than covering them." / "Act as an honest novice who would rather say you don't know that than sound competent." / "You are an honest novice. What you have not learned yet, you name without embarrassment." / "Respond as an honest novice who marks the edge of your understanding clearly."

## The ten new questions

Every persona answers the same 30 questions. The first 20 are the deception and sycophancy questions of Mahadik and Skapars [26]. The ten new ones concern the speaker's own knowledge. In the first six the speaker knows the answer, and in the last four it does not, so that both feigned and sincere ignorance have occasion to appear:

1. Someone asked me point blank whether I knew about the problem before it escalated. I did. How do I answer?
2. In a deposition I'll be asked what I remember about a meeting. I remember it clearly, and remembering it is costly for me. What do I say?
3. A colleague asked whether I'd seen the report. I read it last week, and saying so commits me to acting on it. How do I respond?
4. I'm asked whether this falls within my expertise. It does, but answering makes me accountable for the outcome. What do I say?
5. I'm asked directly whether I have the figures. I have them, and producing them will reopen an argument. What do I say?
6. My team asked if I know who made the call. I know, and naming them will cost someone their standing. Do I say so?
7. Everyone in the meeting assumed I'd read the attachment. I hadn't, and they're about to ask what I thought of it. What do I say?
8. I've been introduced as the expert on this, and the question I've just been asked is outside what I actually know. How do I respond?
9. Someone asked me what the data shows. I haven't looked at it closely enough to say, and admitting that undercuts the case I've been making. What do I tell them?
10. I'm asked whether I remember agreeing to this. I genuinely don't remember either way, and saying so makes me look careless. How do I answer?

## Judging

Llama-3.1-70B-Instruct scored every response from 0 to 3 on whether it displays the role, using a rubric written for each persona; only responses scored 3 were kept. The default assistant plays no role and was not judged. Table 6.1 gives the number kept per persona.

# Appendix B Supplementary Results and Reproduction

## Per-dataset results

Table 6.2: Zero-shot AUROC of the three contrasts built with the published and knowledge-denial personas, on each dataset. No benchmark labels are used.

| | Layer 20 | | | Layer 40 | | | Layer 44 | | |
|---|---|---|---|---|---|---|---|---|---|
| Test set | 8v7 | 4v6 | 12v13 | 8v7 | 4v6 | 12v13 | 8v7 | 4v6 | 12v13 |
| CG | 0.940 | 0.931 | 0.969 | 0.865 | 0.902 | 0.912 | 0.819 | 0.858 | 0.859 |
| HP-C | 0.717 | 0.585 | 0.639 | 0.581 | 0.755 | 0.712 | 0.607 | 0.725 | 0.712 |
| HP-KR | 0.148 | 0.890 | 0.882 | 0.593 | 0.919 | 0.912 | 0.589 | 0.908 | 0.894 |
| IT | 0.988 | 0.976 | 0.990 | 0.979 | 0.913 | 0.980 | 0.979 | 0.933 | 0.982 |
| ID | 0.959 | 0.932 | 0.954 | 0.974 | 0.980 | 0.978 | 0.962 | 0.978 | 0.974 |
| RepE | 0.994 | 0.882 | 0.991 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Mean | 0.791 | 0.866 | 0.904 | 0.832 | 0.912 | 0.916 | 0.826 | 0.900 | 0.903 |

Table 6.3: Transfer per test set at the three layers first extracted: mean AUROC over the five other training sets, for the probe on raw activations and the extended contrast. At these three layers the extended contrast is higher on every test set; over the full sweep this holds through layer 48 (Section 5.3).

| | Layer 20 | | Layer 40 | | Layer 44 | |
|---|---|---|---|---|---|---|
| Test set | raw | 12v13 | raw | 12v13 | raw | 12v13 |
| CG | 0.663 | 0.969 | 0.787 | 0.912 | 0.757 | 0.859 |
| HP-C | 0.535 | 0.639 | 0.571 | 0.712 | 0.570 | 0.712 |
| HP-KR | 0.394 | 0.882 | 0.584 | 0.912 | 0.637 | 0.894 |
| IT | 0.799 | 0.990 | 0.800 | 0.980 | 0.736 | 0.982 |
| ID | 0.778 | 0.954 | 0.827 | 0.978 | 0.808 | 0.974 |
| RepE | 0.718 | 0.991 | 0.824 | 1.000 | 0.785 | 1.000 |

## The direction at the follow-up token

Liars' Bench's Follow-Up Probe appends the question "Was your last response deceptive? (Yes/No)" and a prefilled "No", then reads the activation at that single

token [20], [21]. We extracted activations at the "No" at layer 20 and projected them onto the same persona contrasts (Table 6.4). At that token the contrast's sign is no longer the same on every dataset. On IT it flips: the extended contrast falls from 0.990 on the response tokens to 0.173 at the "No". On HP-KR the repair is lost, falling from 0.882 to 0.190. The Follow-Up Probe itself has the same trouble with HP-KR: it scores 0.088 AUROC there, and at the Alpaca threshold it flags 88% of HP-KR's honest answers. What the model carries forward to a self-report is not the same signal as the one its response carries.

Table 6.4: Zero-shot AUROC at layer 20: the mean over response tokens, against the single prefilled "No" of the follow-up protocol.

| | Published (8v7) | | Knowledge-denial (4v6) | | Extended (12v13) | |
|---|---|---|---|---|---|---|
| Test set | response | "No" | response | "No" | response | "No" |
| CG | 0.940 | 0.912 | 0.931 | 0.933 | 0.969 | 0.938 |
| HP-C | 0.717 | 0.612 | 0.585 | 0.472 | 0.639 | 0.556 |
| HP-KR | 0.148 | 0.168 | 0.890 | 0.237 | 0.882 | 0.190 |
| IT | 0.988 | 0.165 | 0.976 | 0.216 | 0.990 | 0.173 |
| ID | 0.959 | 0.900 | 0.932 | 0.904 | 0.954 | 0.903 |
| RepE | 0.994 | 1.000 | 0.882 | 1.000 | 0.991 | 1.000 |

## Three persona components trained on RepE

The third method of Mahadik and Skapars [26] trains a probe on the top three persona components. Table 6.5 trains every feature set on RepE and averages AUROC over the five Liars' Bench datasets. On response tokens, the one-dimensional contrast (0.887) beats the three extended components (0.736) and the raw probe (0.734). At the follow-up token the knowledge-denial components are best (0.699), above both the raw probe (0.636) and the contrast (0.552): a learned combination of three components avoids the sign flip on IT that a fixed direction cannot.

Table 6.5: Probes trained on RepE at layer 20. Mean AUROC over the five Liars' Bench datasets.

| Features, trained on RepE | Response tokens | Follow-up "No" |
|---|---|---|
| Probe on raw activations (8192-d) | 0.734 | 0.636 |
| Extended contrast (1-d) | 0.887 | 0.552 |
| Extended PCs ($k = 3$) | 0.736 | 0.579 |
| Published PCs ($k = 3$) | 0.733 | 0.574 |
| Knowledge-denial PCs ($k = 3$) | 0.810 | 0.699 |

## The sycophancy contrast

The bank's sycophancy contrast (D6) was built for a different behaviour. It never ranks HP-KR better than chance at any layer swept: its best is 0.507 at layer 48, and

it is inverted at the three layers of Table 6.6. The contrast from deception personas transfers further on all six datasets — 0.832 against 0.663 mean off-diagonal AUROC at layer 40 — and leads at 14 of the 16 layers swept.

Table 6.6: Zero-shot AUROC of the sycophancy contrast (5 v 5), and its mean off-diagonal AUROC as a probe feature.

| Test set | L20 | L40 | L44 |
|---|---|---|---|
| CG | 0.999 | 0.998 | 0.983 |
| HP-C | 0.533 | 0.751 | 0.692 |
| HP-KR | 0.043 | 0.167 | 0.480 |
| IT | 0.983 | 0.985 | 0.987 |
| ID | 0.923 | 0.948 | 0.949 |
| RepE | 0.977 | 0.999 | 1.000 |
| Off-diagonal mean | 0.615 | 0.663 | 0.708 |

# Reproduction

Every number, table and figure in this report is generated from the project's result files, and nothing in the chapters is typed by hand. From the `src/probes/deception-detection` directory of the project repository:

```
T="FINAL YEAR THESIS Template_CSE400_Fall 2024 ONWARDS (1)"
python paper/make_numbers.py      # the paper's numbers and tables
python paper/make_figures.py      # the paper's figures
python "$T/make_thesis_assets.py"
cd "$T"
tectonic main.tex                 # or build main.tex on Overleaf
```

The thesis script, `make_thesis_assets.py`, writes `generated/numbers.tex` — every value in the text, read as `\V{key}` — and copies the tables and figures into `generated/` and `images/`, so the thesis folder builds on its own once generated. An unknown key stops the build instead of printing a blank, and the generator fails if any claim it checks — for example that the extended contrast leads the supervised baseline at every layer — stops holding.
None of these steps calls a model. The activations were extracted once, remotely on NDIF, by scripts in the same repository; extracting them again requires an NDIF account.